United States Patent	12412255
Kind Code	B2
Date of Patent	September 09, 2025
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Enhanced photo relighting based on machine learning models

Abstract

Apparatus and methods related to applying lighting models to images of objects are provided. An example method includes applying a geometry model to an input image to determine a surface orientation map indicative of a distribution of lighting on an object based on a surface geometry. The method further includes applying an environmental light estimation model to the input image to determine a direction of synthetic lighting to be applied to the input image. The method also includes applying, based on the surface orientation map and the direction of synthetic lighting, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the input image. The method additionally includes enhancing, based on the quotient image, a portion of the input image. One or more neural networks can be trained to perform one or more of the aforementioned aspects.

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Appl. No.:	**18/028930**
Filed (or PCT Filed):	**May 17, 2021**
PCT No.:	**PCT/US2021/032734**
PCT Pub. No.:	**WO2022/072000**
PCT Pub. Date:	April 07, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230360182 A1	Nov. 09, 2023

Related U.S. Application Data

us-provisional-application US 63085529 20200930

Publication Classification

Int. Cl.: **G06T5/92** (20240101); **G06T7/60** (20170101); **G06T7/70** (20170101); **G06T15/50** (20110101)

U.S. Cl.:

CPC **G06T5/92** (20240101); **G06T7/60** (20130101); **G06T7/70** (20170101); **G06T15/506** (20130101); G06T2200/24 (20130101); G06T2207/20081 (20130101); G06T2207/20084 (20130101); G06T2207/20092 (20130101); G06T2207/20208 (20130101); G06T2207/30201 (20130101)

Field of Classification Search

CPC: G06T (5/00); G06T (5/92); G06T (7/70); G06T (7/60); G06T (15/506); G06T (2200/24); G06T (2207/20081); G06T (2207/20084); G06T (2207/20092); G06T (2207/20208); G06T (2207/30201); G06T (5/008); G06T (11/001); G06T (11/60); G06K (9/4661); G06N (3/082); G06N (3/042-045); G06V (10/70); G06V (10/75)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS/INCORPORATION BY REFERENCE
(1) This application is a national stage application under 35 U.S.C. § 371 of International Application No. PCT/US2021/032734, filed May 17, 2021, which claims priority to U.S. Provisional Patent Application No. 63/085,529, filed on Sep. 30, 2020, which are hereby incorporated by reference in their entirety.

BACKGROUND
(1) Many modern computing devices, including mobile phones, personal computers, and tablets, include image capture devices, such as still and/or video cameras. The image capture devices can capture images, such as images that include people, animals, landscapes, and/or objects.
(2) Some image capture devices and/or computing devices can correct or otherwise modify captured images. For example, some image capture devices can provide “red-eye” correction that removes artifacts such as red-appearing eyes of people and animals that may be present in images captured using bright lights, such as flash lighting. After a captured image has been corrected, the corrected image can be saved, displayed, transmitted, printed to paper, and/or otherwise utilized.
(3) Professional photographers, such as, for example, portrait photographers, leverage attributes of light on a subject to create compelling photographs of the subject. Such photographers often use specialized equipment, such as off-camera flashes and reflectors, to position lighting and illuminate their subjects to achieve a professional look. In some instances, such activity is performed in controlled studio settings, and involves expert knowledge of the equipment, lighting, and so forth.
(4) Mobile phone users generally don't have access to such specialized portrait studio resources, or knowledge of how to use these resources. However, users may prefer to have access to the professional, high-quality results of seasoned portrait photographers.
SUMMARY
(5) In one aspect, an image capture device may be configured to translate a professional photographer's understanding of light and use of off-camera lighting into a computer-implemented method. Powered by a system of machine-learned components, the image capture device may be configured to enable users to create attractive lighting for portraits or other types of images.
(6) In some aspects, mobile devices may be configured with these features so that an image can be enhanced in real-time. In some instances, an image may be automatically enhanced by the mobile device. In other aspects, mobile phone users can non-destructively enhance an image to match their preference. Also, for example, pre-existing images in a user's image library can be enhanced based on techniques described herein.

(7) In one aspect, a computer-implemented method is provided. A computing device applies the geometry model to an input image to determine a surface orientation map indicative of a distribution of lighting on an object in the input image based on a surface geometry of the object. The computing device applies an environmental light estimation model to the input image to determine a direction of synthetic lighting to be applied to the input image to enhance at least a portion of the input image. The computing device applies, based on the surface orientation map and the direction of synthetic lighting, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the input image. The computing device enhances, based on the quotient image, the portion of the input image.
(8) In another aspect, a computing device is provided. The computing device includes one or more processors and data storage. The data storage has stored thereon computer-executable instructions that, when executed by one or more processors, cause the computing device to carry out functions. The functions include: applying a geometry model to an input image to determine a surface orientation map indicative of a distribution of lighting on an object in the input image based on a surface geometry of the object; applying an environmental light estimation model to the input image to determine a direction of synthetic lighting to be applied to the input image to enhance at least a portion of the input image; applying, based on the surface orientation map and the direction of synthetic lighting, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the input image; and enhancing, based on the quotient image, the portion of the input image.
(9) In another aspect, an article of manufacture is provided. The article of manufacture includes one or more computer readable media having computer-readable instructions stored thereon that, when executed by one or more processors of a computing device, cause the computing device to carry out functions. The functions include: applying a geometry model to an input image to determine a surface orientation map indicative of a distribution of lighting on an object in the input image based on a surface geometry of the object; applying an environmental light estimation model to the input image to determine a direction of synthetic lighting to be applied to the input image to enhance at least a portion of the input image; applying, based on the surface orientation map and the direction of synthetic lighting, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the input image; and enhancing, based on the quotient image, the portion of the input image.
(10) In another aspect, a system is provided. The system includes means for applying a geometry model to an input image to determine a surface orientation map indicative of a distribution of lighting on an object in the input image based on a surface geometry of the object; means for applying an environmental light estimation model to the input image to determine a direction of synthetic lighting to be applied to the input image to enhance at least a portion of the input image; means for applying, based on the surface orientation map and the direction of synthetic lighting, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the input image; and means for enhancing, based on the quotient image, the portion of the input image.
(11) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the figures and the following detailed description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) FIG. **1** illustrates images **100** with different lighting, in accordance with example embodiments.

(2) FIG. **2** is a diagram depicting a relighting network for enhancing lighting of images, in accordance with example embodiments.
(3) FIG. **3** is a diagram depicting an example network to predict a surface orientation map, in accordance with example embodiments.
(4) FIG. **4** is an example architecture for a neural network to predict a surface orientation map, in accordance with example embodiments.
(5) FIG. **5** illustrates example images from a photographed reflectance field, in accordance with example embodiments.
(6) FIG. **6** illustrates an example method for recommending a light direction, in accordance with example embodiments.
(7) FIG. **7** illustrates example images with estimated lighting, in accordance with example embodiments.
(8) FIG. **8**A illustrates an example network architecture to determine a light visibility map, in accordance with example embodiments.
(9) FIG. **8**B illustrates an example image highlighting an effect of surface geometry in photo illumination, in accordance with example embodiments.
(10) FIG. **9** is a diagram depicting an example network to predict a quotient image, in accordance with example embodiments.
(11) FIG. **10** is an example architecture for a neural network to predict a quotient image, in accordance with example embodiments.
(12) FIG. **11** is an example network illustrating post-processing techniques, in accordance with example embodiments.
(13) FIG. **12** is an example interactive graphical user interface, in accordance with example embodiments.
(14) FIG. **13** illustrates an example input-output image pair based on a surface orientation map, in accordance with example embodiments.
(15) FIG. **14** illustrates an example relighting pipeline, in accordance with example embodiments.
(16) FIG. **15** illustrates example input images with predicted face light estimations, in accordance with example embodiments.
(17) FIG. **16** is a diagram illustrating training and inference phases of a machine learning model, in accordance with example embodiments.
(18) FIG. **17** depicts a distributed computing architecture, in accordance with example embodiments.
(19) FIG. **18** is a block diagram of a computing device, in accordance with example embodiments.
(20) FIG. **19** depicts a network of computing clusters arranged as a cloud-based server system, in accordance with example embodiments.
(21) FIG. **20** is a flowchart of a method, in accordance with example embodiments.

DETAILED DESCRIPTION

(22) This application relates to enhancing an image of an object, such as an object depicting a human face, using machine learning techniques, such as but not limited to neural network techniques. When a mobile computing device user takes an image of an object, such as a person, the resulting image may not always have ideal lighting. For example, the image could be too bright or too dark, the light may come from an undesirable direction, or the lighting may include different colors that give an undesirable tint to the image. Further, even if the image does have a desired lighting at one time, the user might want to change the lighting at a later time. As such, an image-processing-related technical problem arises that involves adjusting lighting of an already-obtained image.
(23) To allow user control of lighting of images, particularly images of human faces, the herein-described techniques apply a model based on a convolutional neural network to adjust lighting of images. The herein-described techniques include receiving an input image and data about a

particular lighting model to be applied to the input image, predicting an output image that applies the data about the particular lighting model to be applied to the input image using the convolutional neural network, and generating an output based on the output image. The input and output images can be high-resolution images, such as multi-megapixel sizes images captured by a camera of a mobile computing device. The convolutional neural network can work well with input images captured under a variety of natural and artificial lighting conditions. In some examples, a trained model of the convolutional neural network can work on a variety of computing devices, including but not limited to, mobile computing devices (e.g., smart phones, tablet computers, cell phones, laptop computers), stationary computing devices (e.g., desktop computers), and server computing devices. The convolutional neural network can apply the particular lighting model to an input image, thereby adjusting the lighting of the input image and solving the technical problem of adjusting the lighting of an already-obtained image.

(24) A neural network, such as a convolutional neural network, can be trained using a training data set of images to perform one or more aspects as described herein. In some examples, the neural network can be arranged as an encoder/decoder neural network.

(25) While examples described herein relate to determining and applying lighting models of images of objects with human faces, the neural network can be trained to determine and apply lighting models to images of other objects, such as objects that reflect light similarly to human faces. Human faces typically reflect light diffusely but can also include some specular highlights due to directly reflected light. For example, specular highlights can result from direct light reflections from eye surfaces, glasses, jewelry, etc. In many images of human faces, such specular highlights are relatively small in area in proportion to areas of facial surfaces that diffusely reflect light. Thus, the neural network can be trained to apply lighting models to images of other objects that diffusely reflect light, where these diffusely reflecting objects may have some relatively-small specular highlights (e.g., a tomato or a wall painted with matte-finish paint). The images in the training data set can show one or more particular objects using lighting provided under a plurality of different conditions, such as lighting provided from different directions, lighting provided of varying intensities (e.g. brighter and dimmer lighting), lighting provided with light sources of different colors, lighting provided with different numbers of light sources, etc.

(26) A trained neural network can process the input image to predict the environmental illumination. An optimal light direction, that complements the existing portrait lighting of the input image, can be recommended based on the predicted environmental illumination. The trained relighting network can take the optimal light direction, a prediction of a surface geometry of an object in the input image, and a prediction of a quotient image indicative of an amount of light energy to be applied to each pixel of the input image. The trained neural network can also process the image to apply the desired lighting to the original image and predict an output image where the desired lighting has been applied to the input image from the recommended light direction. Then, the trained neural network can provide outputs that include the predicted output image.

(27) In one example, (a copy of) the trained neural network can reside on a mobile computing device. The mobile computing device can include a camera that can capture an input image of an object, such as a portrait of a person's face. A user of the mobile computing device can view the input image and determine that the input image should be relit. The user can then provide the input image and the information on how the input image should be relit to the trained neural network residing on the mobile computing device. In response, the trained neural network can generate a predicted output image that shows the input image relit as indicated by the user and subsequently output the output image (e.g., provide the output image for display by the mobile computing device). In other examples, the trained neural network is not resident on the mobile computing device; rather, the mobile computing device provides the input image and the information on how the input image should be relit to a remotely-located trained neural network (e.g., via the Internet or another data network). The remotely-located convolutional neural network can process the input

image and the information on how the input image should be relit as indicated above and provide an output image that shows the input image relit as indicated by the user to the mobile computing device. In other examples, non-mobile computing devices can also use the trained neural network to relight images, including images that are not captured by a camera of the computing device.
(28) In some examples, the trained neural network can work in conjunction with other neural networks (or other software) and/or be trained to recognize whether an input image of an object is poorly lit. Then, upon a determination that an input image is poorly lit, the herein-described trained neural network could apply a corrective lighting model to the poorly-lit input image, thereby correcting the poor lighting of the input image. The corrective lighting model can be chosen based on user input and/or be predetermined. For example, a user input lighting model or a predetermined lighting model can be used to provide a “flat light” or light resembling the technique of “fill flash”, or light the object with a lighting model that raises undesirable shadows on a face (e.g., from a backlit scene) with respect to the facial geometry, thus correcting flattening of the image.
(29) In some examples, the trained neural network can take as inputs one input image and one or more lighting models and provide one or more resulting output images. Then, the trained neural network can determine the one or more resulting output images by applying each of the plurality of the lighting models to the input image. For example, the one or more lighting models can include a plurality of lighting models that represent one (or more) light source(s) that change location, lighting color, and/or other characteristics in each of the plurality of lighting models. More particularly, the plurality of lighting models could represent one or more light sources, where at least one light source changes location (e.g., by a predetermined amount) between provided models. In this approach, the resulting output images represent the input image shown as the changing light source(s) appear(s) to rotate or otherwise move about an object (or objects) depicted in the input image. Similarly, the changing light source(s) could change color (e.g., by a predetermined distance in a color space) between provided models so that the resulting output images represent the input image shown with a variety of colors of light. The plurality of output images could be provided as still images and/or as video imagery. Other effects could be generated by having the trained neural network apply a plurality of lighting models to one image (or relatedly, having the trained neural network apply one lighting model to a plurality of input images).
(30) As such, the herein-described techniques can improve images by applying more desirable and/or selectable lighting models to images, thereby enhancing their actual and/or perceived quality. Enhancing the actual and/or perceived quality of images, including portraits of people, can provide emotional benefits to those who believe their pictures look better. These techniques are flexible, and so can apply a wide variety of lighting models to images of human faces and other objects, particularly other objects with similar lighting characteristics. Also, by changing a lighting model, different aspects of an image can be highlighted which can lead to better understanding of the object(s) portrayed in the image.
(31) Techniques for Image Relighting Using Neural Networks
(32) FIG. **1** illustrates images **100** with different lighting, in accordance with example embodiments. Images **100** include image **110** and image **120** depicting a human face of the same person under two different types of lighting. Image **110** is relit in accordance with example embodiments to obtain image **130**, where the right side of the face in image **110** is illuminated. Image **130** is an image that includes a human face, where the lighting is dimmer than the lighting in image **140**. Other examples of images with different lighting and/or other types of imperfect lighting are possible as well. Images **100** illustrate an impact of lighting on an image. For example, lighting can impact a perceived expression, a perceived mood, and/or other aspects of a subject's features and/or personality.
(33) Relighting Network
(34) A relighting network can be designed to computationally add an additional, repositionable light source into an image, with an initial lighting direction and an intensity automatically selected

to complement original lighting conditions of the image. For example, under less-than-ideal original lighting conditions, for instance in backlit scenes, this additional light source improves exposure on the eyes and face. If the image already has compelling lighting, as is often the case for scenes with some directional illumination, the image can be gracefully enhanced for dramatic effect, for example, by accentuating contouring and shaping of a face or a person in the image.

(35) FIG. **2** is a diagram depicting a relighting network **200** for enhancing lighting of images, in accordance with example embodiments. An input image **210** may be utilized by a first convolutional neural network **215** to generate a surface orientation map **220** indicative of a surface geometry of input image **200**. Generally, when photographers add an additional light source into a scene, the orientation of the light source relative to the subject's facial geometry determines how much brighter each part of the face appears in the input image **210**. In optics, based on Lambert's cosine law, an object composed of a relatively matte material reflects an amount of light proportional to the cosine of the angle between its surface orientation and the direction of the incident light. To model this behavior, the first convolutional neural network **215** is trained to estimate surface orientation from the input image **210**. The output of the first convolutional neural network **215** is a colorized representation of a collection of 3-dimensional vectors in camera space.

(36) For example, the spatial vector can be mapped to a colorized red-green-blue (RGB) surface orientation map **220**. As illustrated, an original light source is illuminating a left side of the face in input image **210**, there is less illumination on the right side of the face in input image **210**, and the background and hair color are of a substantially darker color. In surface orientation map **220**, the illuminated left side of the face corresponds to blue colored pixels, the background and hair color correspond to green colored pixels, and the less illuminated right side of the face corresponds to red colored pixels.

(37) In some aspects, a second convolutional neural network **225** is trained to estimate the environmental lighting corresponding to the input image **210**. For example, in order to recommend an optimal light direction, a second convolutional neural network **225** is trained to estimate a high dynamic range, omnidirectional illumination profile for a scene based on an input portrait. This lighting estimation model can infer the direction, relative intensity, and color of all light sources coming from all directions, in the scene in input image **210**, by considering the face to be a light probe. The environment light estimation is then utilized to automatically determine an optimal light direction **230** for the relighting network **200**. In some aspects, a pose of a portrait's subject may be estimated to determine an optimal light direction **230**. Also, for example, labeled image data may be generated that associates different positions of light sources with optimal light placements for images, and a machine learning model may be trained on the labeled data to automatically determine light direction **230**. For example, in studio portrait photography, a main off-camera light source, or “key-light” is often placed about 30° above the eye-line of a subject, and between 30° and 60° off the camera axis when looking overhead at the subject. Relighting network **200** can be configured to follow a similar guideline for a classical portrait look, thereby enhancing pre-existing lighting directionality in input image **210** while targeting a balanced, subtle key-to-fill lighting ratio of about 2:1.

(38) As previously described, input image **210** indicates that the right side of the face is less illuminated than the left side of the face. Accordingly, in some embodiments, relighting network **200** can automatically determine light direction **230** to illuminate this less illuminated right side of the face. In some embodiments, relighting network **200** can receive a user preference of a light direction **230** via an interactive graphical user interface.

(39) Based on Lambert's cosine law, a dot product **235** may be computed between a three dimensional vector representation of surface orientation map **220** and a three dimensional vector representation of light direction **230**, to generate light visibility map **240**. Generally, light visibility map **240** is indicative of regions in a portrait that are to be illuminated via synthetic lighting. For example, light visibility map **240** indicates portions of input image **200** where light can be seen and

where light cannot be seen based on surface orientation. As previously described, input image **210** indicates that the right side of the face is less illuminated than the left side of the face, and relighting network **200** can automatically determine light direction **230** to illuminate this less illuminated right side of the face. Also, for example, surface orientation map **220** takes into account a surface geometry of the face in input image **200** to highlight portions, in red colored pixels, where environmental lighting does not provide adequate illumination. Accordingly, based on light direction **230** and surface orientation map **220**, light visibility map **240** indicates the right side of the face as the portion that requires illumination.

(40) When enhancing input image **210** in near real-time, it is beneficial to decrease utilization of computing resources, including for example, processing time, processing speed, and memory allocation. Specifically, it may not be desirable to directly utilize light visibility map **240** to enhance input image **210**. The light visibility map **240** indicates areas of input image **210** that need to be illuminated and how much of the light illuminates these areas, but light visibility map **240** does not capture material properties of the object that is being relit. Depending on the material of the object being relit, the same light may lead to very different results e.g. shiny/specular materials vs dull/diffuse materials reflect light very differently.

(41) Instead, light visibility map **240** and input image **210** can be input to third convolutional neural network **245**, to predict a quotient image **250**. The quotient prediction network **245** has to learn properties of skin, eyes, even materials on clothing from training data, and produce a quotient image **250** that takes into account both the material properties learned from the input image **210**, and the optimized light direction and geometry information provided by the light visibility map **240**. Generally, quotient image **250** is a per pixel real-valued, multiplicative factor indicating an amount of illumination to be applied to each pixel of input image **210**. This may be further supervised with a ground truth image. Because quotient image **250** is a multiplicative factor, it does not dampen details from the original input image **210**. For example, even at a much lower resolution, such as for blurry images, quotient image **250** can be predicted. On the other hand, processing the high resolution input image **210** may involve more computational complexity, causing delays in real-time processing. Generally, details in a high resolution input image **210** can be preserved, while enhancing input image **210** to bring out less visible aspects, thereby outputting a realistic image. The multiplier in quotient image **250** makes a given pixel lighter or darker. The range of values may vary, sometimes 10×, depending on an intensity of the synthetic illumination. Quotient image **250** enhances low-frequency lighting changes, without impacting high-frequency image details, which are directly transferred from input image **210** to maintain image quality. In some aspects, post-processing can be applied to quotient image **250** to adjust highlighting, exposure adjustment, and matting, thereby rendering a photorealistic image enhancement.

(42) Multiplication **255** of quotient image **250** and input image **210** generates relit image **260**. This method is also computationally efficient, as third convolutional neural network **245** predicts a lower-resolution quotient image that is upsampled prior to multiplication **255** with the high-resolution input. Accordingly, relighting network **200** combines surface geometry, and an automatic estimation of light direction, to generate relit image **260**, where high frequency details of input image **210** are preserved, and low frequency details of input image **210** are enhanced. In some aspects, relighting network **200** can be optimized to run at interactive frame-rates on mobile devices, with a total model size under 10 MB. The results can be produced with a version of a UNet model that utilizes a combination of standard and separable, depth-wise convolutions, and concatenations for skip connections. This, along with float **16** quantization, can lead to a model size of 2.4 MB per UNet, and thus a total model size of 4.8 MB.

(43) An interactive user experience may be provided where a user can adjust light position and brightness. This provides additional creative flexibility to users to find their own balance between light and shadow. Also, for example, aspects of relighting network **200** can be applied to existing images from a user's photo library. For example, for existing images where faces may be slightly

underexposed, relighting network **200** can be applied to illuminate and shape the face. This may especially benefit images with a single individual posed directly in front of the camera.

(44) Several aspects of relighting network **200** are described in greater detail below. For example, relighting network **200** can take an input image **210**, and without any prior knowledge of cameras, exposures, compositions, and without any additional photographic hardware equipment, relighting network **200** can derive the geometry, lighting, and find an optimal exposure to enhance input image **210**. Based on an estimation of original lighting in the input image **210**, relighting network **200** can automatically deduce an optimal light direction for synthetic lighting. Combining the optimal light direction with a knowledge of surface geometry customizes an application of the synthetic lighting to geometric features of an object in input image **210**. Also, for example, using a per pixel multiplicative factor to maintain high frequency details of the input image, while enhancing low frequency lighting details, is a significant factor in reducing computational complexity of image enhancement techniques. Further post-processing techniques to adjust highlighting, exposure adjustment, and matting, enable rendering of a photorealistic image enhancement. Intermediate interpretable outputs, such as, for example, surface orientation map **220**, light direction **230**, light visibility map **240**, and quotient image **250**, provide opportunities to optimize loss functions for neural networks, improve image quality, and otherwise make intermediate adjustments that inform the overall quality of the enhanced image. For example, surface orientation map **220** can be utilized to identify sources of error, and ground truth data may be updated with additional input from photographic techniques (e.g., adjustments made by professional photographers) to correct the errors. An interactive user experience, where a user can adjust light position and brightness for any image, is another significant feature of techniques described herein.

(45) Surface Orientation Map—Supervising Relighting with Geometry Estimation

(46) FIG. **3** is a representation of an example network to predict a surface orientation map, in accordance with example embodiments. For example, a convolutional neural network **215** may take input image **210**, infer a reflectance field **310** indicative of a surface geometry of an object in input image **210** (e.g., a face in a portrait, and/or an entire body in the input image **210**). Based on ground truth data indicative of correlations between lighting and surface geometry, convolutional neural network **215** can predict surface orientation map **220**. As described herein, surface orientation map **220** can be combined with one or more aspects of relighting network **200** to generate relit image **260**. Also, for example, additional or alternative methods for ground truth geometry/surface orientation can be applied, such as, for example, data from depth sensors. In some example embodiments, depth from a mobile phone depth sensor can be used to determine a surface orientation map **220**, instead of using a network (e.g., convolutional neural network **215**).

(47) A four-dimensional reflectance field **310**, R(u, v, θ, ϕ), may represent a subject lit from any lighting direction (θ, ϕ) for each image pixel(u, v), according to the light data. Generally, reflectance field **310** describes how a volume of space enclosed by a surface A transforms a directional illumination (θ.sub.i, ϕ.sub.i) into a radiant field of illumination R(θ.sub.r, ϕ.sub.r, u.sub.r, v.sub.r) at a point (u.sub.r, v.sub.r). The light data represents one of a specified number of directions from which a face is illuminated for a portrait used in the image training data to train convolutional neural network **215**.

(48) While examples described herein relate to determining and applying lighting models of images of objects with human faces, convolutional neural network **215** can be trained to determine and apply lighting models to images of other objects, such as objects that reflect light similarly to human faces. Human faces typically reflect light diffusely but may include some specular highlights due to directly reflected light. For example, specular highlights can result from direct light reflections from eye surfaces, glasses, jewelry, etc. In many images of human faces, such specular highlights are relatively small in area in proportion to areas of facial surfaces that diffusely reflect light. Thus, convolutional neural network **215** can be trained to apply lighting models to

images of other objects that diffusely reflect light, where these diffusely reflecting objects may have some relatively-small specular highlights (e.g., a tomato or a wall painted with matte-finish paint). The images in the training data set can show one or more particular objects using lighting provided under a plurality of different conditions, such as lighting provided from different directions, lighting provided of varying intensities (e.g. brighter and dimmer lighting), lighting provided with light sources of different colors, lighting provided with different numbers of light sources, etc. Once trained, convolutional neural network **215** can receive an input image **210** and information about original lighting. The trained convolutional neural network **215** can process the input image **210** to determine a prediction of lighting based on surface geometry, thereby generating surface orientation map **220**.

(49) FIG. **4** is an example architecture for a neural network to predict a surface orientation map, in accordance with example embodiments. For example, convolutional neural network **215** can be modeled around a UNet with skip connections. In some embodiments, the input image **210** can be of size (256×192×3) and can go through 6 encoder blocks, Enc L1 410, Enc L2 **412**, Enc L3 **414**, Enc L4 **416**, Enc L5 **418**, and Enc L6 **420**, and 6 decoder blocks, Dec L1 **422**, Dec L2 **424**, Dec L3 **426**, Dec L4 **428**, Dec L5 **430**, and Dec L6 **432**. Each encoder block of encoder **401** consists of a single convolution followed by downsampling by a factor of 2 followed by a blur-pooling operation. Each decoder block of decoder **403** consists of a single convolution followed by upsampling by a factor of 2 followed by a bilinear upsampling operation. Skip connections from the encoder **401** are concatenated or added to the upsampled features of decoder **403** depending on the network version. In some example embodiments, filters for encoder **401** may include (16, 32, 64, 128, 256, 256) filters for the encoder blocks yielding a final bottleneck size of (4×3×256). Similarly, filters for decoder **403** may include (256, 256, 128, 64, 32, 16) features to return back to the (256×192) image resolution. A final convolution with 3 filters can be used to produce surface orientation map **220** from the decoder output. Relu activations can be used after all convolutions.

(50) The image training data **405** represents a set of portraits of faces photographed with various lighting arrangements. In some implementations, the image training data **405** includes images of faces, or portraits, formed with high-dynamic range (HDR) illumination recovered from low-dynamic range (LDR) lighting environment capture. As shown in FIG. **4**, the image training data **405** includes multiple images **406**(**1**), . . . **406**(M), where M is the number of images in the image training data **405**. Each image, such as, image **406**(**1**) includes light data **408**(**1**) and may also include pose data.

(51) The light data **408**(**1** . . . M), based on ground truth data collection, represents one of a specified number (e.g., 331) of directions from which a face is illuminated for a portrait used in the image training data **405**. In some implementations, the light data **408**(**1**) includes a polar angle and an azimuthal angle, i.e., coordinates on a unit sphere. In some implementations, the light data **408**(**1**) includes a triplet of direction cosines. In some implementations, the light data **408**(**1**) includes a set of Euler angles. In some implementations, the angular configuration represented by the light data **408**(**1**) is one of **331** configurations used to train convolutional neural network **215**.

(52) In some implementations, to photograph a subject's reflectance field, a computer-controllable sphere of white LED light sources can be used with lights spaced 120 apart at the equator. In such implementations, the reflectance field is formed from a set of reflectance basis images, photographing the subject as each of the directional LED light sources is individually turned on one-at-a-time within the spherical rig. Such One-Light-at-a-Time (OLAT) images are captured for multiple camera viewpoints. In some implementations, the 331 OLAT images are captured for each subject using six color machine vision cameras with 12-megapixel resolution, placed 1.7 meters from the subject, although these values and number of OLAT images and types of cameras used may differ in some implementations. In some implementations, cameras are positioned roughly in front of the subject, with five cameras with 35 mm lenses capturing the upper body of the subject from different angles, and one additional camera with a 50 mm lens capturing a close-up image of

the face with tighter framing.
(53) In some implementations, reflectance fields for 70 diverse subjects are used, each subject performing ten different facial expressions and wearing different accessories, yielding about 700 sets of OLAT sequences from six different camera viewpoints, for a total of 4200 unique OLAT sequences. Other quantities of sets of OLAT sequences may be used. Subjects spanning a wide range of skin pigmentations were photographed. Also, for example, 32 custom high resolution (e.g., 12 MP) depth sensors may be used. As another example, 62 high resolution (e.g., 12 MP) RGB cameras can be used. Together, over 15 million images may be generated as image training data **405**.
(54) As acquiring a full OLAT sequence for a subject takes some time, e.g., around six seconds, there may be some slight subject motion from frame-to-frame. In some implementations, an optical flow technique is used to align the images, interspersing occasionally (e.g., at every 11th OLAT frame) one extra "tracking" frame with even, consistent illumination to ensure the brightness constancy constraint for optical flow is met. This step may preserve the sharpness of image features when performing the relighting operation, which linearly combines aligned OLAT images.
(55) FIG. **5** illustrates example images from a photographed reflectance field, in accordance with example embodiments. For example, a spherical lighting rig can include 64 cameras with different viewpoints and **331** individually-programmable LED light sources. As illustrated in FIG. **5**, each individual can be photographed as illuminated OLAT by each light, forming a reflectance field (e.g., reflectance field **310** of FIG. **3**). Images **510** depict an individual's appearance as illuminated by tiny cones of the **3600** environment. The reflectance field **310** encodes unique colors and light-reflecting properties of each individual's skin, hair, and clothing. For example, reflectance field **310** encodes information on how shiny or dull each material appears. Due to the superposition principle for light, these OLAT images can then be linearly added together to render realistic images **520** of the individual as they would appear in any image-based lighting environment, with complex light transport phenomena like subsurface scattering correctly represented. Synthetic portraits of each individual may be generated in many different lighting environments both with and without added directional light, rendering millions of pairs of images for image training data **405**. The dataset can include model performance across diverse lighting environments and individuals.
(56) Convolutional neural network **215** can be a fully-convolutional neural network as described herein. During training, convolutional neural network **215** can receive as inputs one or more input training images. Convolutional neural network **215** can include layers of nodes for processing input image **210**. Example layers can include, but are not limited to, input layers, convolutional layers, activation layers, pooling layers, and output layers. Input layers can store input data, such as pixel data of input image **210** and inputs from other layers of convolutional neural network **215**. Convolutional layers can compute an output of neurons connected to local regions in the input. In some examples, the predicted outputs can be fed back into the convolutional neural network **215** again as input to perform iterative refinement. Activation layers can determine whether or not an output of a preceding layer is "activated" or actually provided (e.g., provided to a succeeding layer). Pooling layers can downsample the input. For example, convolutional neural network **215** can involve one or more pooling layers downsample the input by a predetermined factor (e.g., a factor of two) in the horizontal and/or vertical dimensions. Output layers can provide an output of conventional neural network **215** to software and/or hardware interfacing with conventional neural network **215**; e.g. to hardware and/or software used to display, print, communicate and/or otherwise provide surface orientation map **220** (e.g., to one or more components of relighting network **200**). Layers **410**, **412**, **414**, **416**, **418**, **420**, **422**, **424**, **426**, **428**, **430**, **432** can include one or more input layers, output layers, convolutional layers, activation layers, pooling layers, and/or other layers described herein.
(57) In some implementations, convolutional neural network **215** can include encoding layers **410**, **412**, **414**, **416**, **418** arranged respectively as in an order as layers L1, L2, L3, L4, L5 each

successively convolving its input and providing its output to a successive layer until reaching encoding layer L6 **420**. In FIG. **4**, a depicted layer can include one or more actual layers. For example, encoding layer L1 **410** can have one or more input layers, one or more activation layers, and/or one or more additional layers. As another example, encoding layer L2 **412**, encoding layer L3 **414**, encoding layer L4 **416**, encoding layer L5 **418**, and/or encoding layer L6 **420** can include one or more convolutional layers, one or more activation layers (e.g., having a one-to-one relationship to the one or more convolutional layer), one or more pooling layers, and/or one or more additional layers.
(58) In some examples, some or all of the pooling layers in convolutional neural network **215** can downsample an input by a common factor in both horizontal and vertical dimensions, while not downsampling depth dimensions associated with the input. The depth dimensions could store data for pixel colors (red, green, blue) and/or data representing scores. Other common factors for downsampling other than two can be used as well by one or more (pooling) layers of convolutional neural network **215**.
(59) Encoding layer L1 **410** can receive and process input image **210** and provide an output to encoding layer L2 **412**. Encoding layer L2 **412** can process the output of encoding layer L1 **410** and provide an output to encoding layer L3 **414**. Encoding layer L3 **414** can process the output of encoding layer L2 **412** and provide an output to encoding layer L4 **416**. Encoding layer L4 **416** can process the output of encoding layer L3 **414** and provide an output to encoding layer L5 **418**. Encoding layer L5 **418** can process the output of encoding layer L4 **416** and provide an output to encoding layer L6 **420**.
(60) Encoding layer L6 **420** may provide the output to decoding layer L1 **422** to begin predicting surface orientation map **220**. Decoding layer L2 **424** can receive and process inputs from both decoding layer L1 **422** and encoding layer L5 **418** (e.g., using a skip connection between encoding layer L5 **418** and decoding layer L2 **424**) to provide an output to decoding layer L3 **426**. Decoding layer L3 **426** can receive and process inputs from both decoding layer L2 **424** and encoding layer L4 **416** (e.g., using a skip connection between encoding layer L4 **416** and decoding layer L3 **426**) to provide an output to decoding layer L4 **428**. Decoding layer L4 **428** can receive and process inputs from both decoding layer L3 **426** and encoding layer L3 **414** (e.g., using a skip connection between encoding layer L3 **414** and decoding layer L4 **428**) to provide an output to decoding layer L5 **430**. Decoding layer L5 **430** can receive and process inputs from both decoding layer L4 **428** and encoding layer L2 **412** (e.g., using a skip connection between encoding layer L2 **412** and decoding layer L5 **430**) to provide an output to decoding layer L6 **432**. Decoding layer L6 **432** can receive and process inputs from both decoding layer L5 **430** and encoding layer L1 **410** (e.g., using a skip connection between encoding layer L1 **410** and decoding layer L6 **432**) to provide a prediction of surface orientation map **220**, which can then be output from decoding layer L6 **432**. The data provided by skip connections between encoding layers **418**, **416**, **414**, **412**, **410** and respective decoding layers **424**, **426**, **428**, **430**, **432** can be used by each respective decoding layer to provide additional details for generating a decoding layer's contribution to the prediction of surface orientation map **220**. In some examples, each of decoding layers **422**, **424**, **426**, **428**, **430**, **432** used to predict surface orientation map **220** can include one or more convolution layers, one or more activation layers, and perhaps one or more input and/or output layers. In some examples, some or all of layers **410**, **412**, **414**, **416**, **418**, **420**, **422**, **424**, **426**, **428**, **430**, **432** can act as a convolutional encoder/decoder network.
(61) In some implementations, convolutional neural network **215** is trained end to end with losses on the predicted surface orientation map **220**. For example, a combination of L1 and adversarial losses can be used to train convolutional neural network **215**. Generally, empirical data suggests that adversarial loss is a significant factor for good generalization to images in the wild when training data is limited (e.g., 15 subjects). However, L1 loss achieved similar results with a larger dataset (e.g., 70 subjects). Furthermore, adversarial loss can become harder to train with a larger

variation in viewpoints and subject clothing. In some implementations, adversarial loss may be selectively applied to the face portion of an image. Other loss measures can be used as well or instead. For example, an L2 loss measure between surface orientation map predictions and training images can be minimized during training of convolutional neural network **215** for predicting surface orientation map **220**.

(62) As described herein, convolutional neural network **215** can include perceptual loss processing. For example, convolutional neural network **215** can use generative adversarial net (GAN) loss functions to determine if part or all of an image would be predicted to generate a surface orientation map, and so satisfy one or more perceptually-related conditions on lighting of that part of the image. In some examples, cycle loss can be used to feed predicted surface orientation maps back into convolutional neural network **215** to generate and/or refine further predicted surface orientation maps. In some examples, convolutional neural network **215** can utilize deep supervision techniques to provide constraints on intermediate layers. In some examples, convolutional neural network **215** can have more, fewer, and/or different layers to those shown in FIG. **4**.

(63) Light Direction—Automatic Light Placement

(64) FIG. **6** illustrates an example method for recommending a light direction, in accordance with example embodiments. Input image **210** can be processed by a face detection component **610** to detect a human face in input image **210**. Image **620** illustrates the face of a human subject within a bounding box. This portion of image **620** can be cropped and resized (e.g., enlarged or shrunk) to obtain face **630**. Although background imagery may provide contextual cues that aid in lighting estimation, some implementations compute a face bounding box for each input, and during training and inference some implementations crop each image, expanding the bounding box by 25%. During training some implementations add slight crop region variations, randomly changing their position and extent.

(65) Face **630** and/or image **620** may be an input to convolutional neural network **225**. In some embodiments, the architectural components of convolutional neural network **225** may be similar to those for convolutional neural network **215** described with respect to FIG. **4**. Convolutional neural network **225** may take input face **630** and predict the environmental illumination profile for input image **210**. Convolutional neural network **225** is a lighting estimation model that infers the direction, relative intensity, and color of all light sources in face **630** coming from all directions, considering face **630** as a light probe.

(66) For example, an image of a person can be taken while being lighted by each individual light source in image **620** and/or face **630**. In some aspects, training data for convolutional neural network **225** may include image training data **405** of FIG. **4**. During training, convolutional neural network **225** can receive as inputs one or more input training images and input target lighting profiles. For example, convolutional neural network **225** can be trained on input image **410**. During training, convolutional neural network **225** is being directed to generate predictions of original lighting profiles as estimated lighting **640**.

(67) Original lighting model is a light estimation model that predicts the actual lighting conditions used to illuminate input image **210**. A lighting model can include a grid or other arrangement of lighting profile data related to lighting of part or all of one or more images. The lighting-profile data can include, but is not limited to, data representing one or more colors, intensities, directions, for the lighting of part or all of the one or more images. Convolutional neural network **225** can be configured to produce a predicted illumination profile or estimated lighting **640** based on the plurality of images of the plurality of human faces. The convolutional neural network **225** is configured to produce a predicted illumination profile or estimated lighting **640** based on input image **210**. The input image **210** represents at least one human face. In some aspects, the convolutional neural network **225** includes a cost function (e.g., a discriminator and cost function data) that is based on a plurality of bidirectional reflectance distribution functions (BRDFs) corresponding to each of the reference objects. Estimated lighting **640** represents a spatial

distribution of illumination incident on a subject of a portrait. An example representation of estimated lighting **640** includes coefficients of a spherical harmonic expansion of a lighting function of angle. Another example representation of estimated lighting **640** includes a grid of pixels, each having a value of the lighting function of solid angle.

(68) As described herein, the input to the convolutional neural network **225** is an sRGB encoded LDR image, e.g., an LDR portrait **620**, with the crop **630** of the face region of each image detected by face detector **610**, resized to an input resolution of 256×256, and normalized to the range of [−0.5,0.5]. Convolutional neural network **225** has an encoder/decoder architecture including the with a latent vector representation of log-space HDR illumination of size **1024** at the bottleneck. In some implementations, the encoder and the decoder are implemented as convolutional neural networks. In some implementations, the encoder includes five 3×3 convolutions each followed by a blur-pooling operation, with successive filter depths of 16, 32, 64, 128, and 256, followed by one last convolution with a filter size of 8×8 and depth **256**, and finally a fully-connected layer. The decoder includes three sets of 3×3 convolutions of filter depths **64**, **32**, and **16**, each followed by a bilinear upsampling operation. The final output of the convolutional neural network **225** includes a 32×32 HDR image of estimated lighting **640**, which can be visualized by rendering three different spheres (diffuse, matte silver, and mirror), and representing log-space omnidirectional illumination.

(69) FIG. **7** illustrates example images with estimated lighting, in accordance with example embodiments. For example, images **710**, **720** and **730** are shown with respective estimated lighting **640**A, **640**B and **640**C. For example, estimated lighting **640**A, **640**B and **640**C estimate the high dynamic range, omnidirectional illumination profile from image **710**, **720** and **730** respectively. The three spheres (**640**A, **640**B and **640**C) at the right of each image are rendered using the estimated illumination, showing the color, intensity, and directionality of the environmental lighting in images **710**, **720** and **730** respectively.

(70) Referring back to FIG. **6**, input image **210** can be processed to estimate pose **650** for a human in input image **210**. In some embodiments, this operation may be performed by a face geometry solution, such as a face mesh. A machine learning model can infer the 3D surface geometry with only a single camera input without a dedicated depth sensor. Face geometry data can consist of common 3D geometry primitives, including a face pose transformation matrix and a triangular face mesh. Also, for example, a machine learning solution for high-fidelity upper-body pose tracking, inferring **25** 2D upper-body landmarks from RGB video frames, can be configured. Using a detector, an estimated pose detector first locates the pose region-of-interest (ROI) within an image frame. The detector subsequently predicts the pose landmarks within the ROI using the ROI-cropped frame as input.

(71) Photographers often rely on perceptual cues when deciding how to augment environmental illumination with off-camera light sources. They assess the intensity and directionality of the light falling on the face, and also adjust their subject's head pose to complement it. The computational equivalents of these operations are estimated lighting **640** and estimated pose **650**. Also, for example, in studio portrait photography, the main off-camera light source, or “key-light” is typically placed about 30° above the eye-line of a subject, and between 30° and 60° off the camera axis when looking overhead at the subject. These guidelines can be configured to be performed computationally based on estimated lighting **640** and estimated pose **650** to automatically suggest an optimal light direction **230**. For example, light direction **230** indicates a direction from which synthetic lighting may be added to input image **210**, similar to how a professional photographer can place an off-camera light source to illuminate a subject. Thus, pre-existing lighting directionality in the scene can be enhanced while targeting a balanced, subtle key-to-fill lighting ratio of about 2:1. As used herein, fill lighting generally refers to additional lighting that may be added to illuminate low frequency portions of an image.

(72) Given a desired lighting direction **230** and input image **210**, a machine learning model can be trained to add the synthetic illumination from a directional light source to an original photograph.

To supervise training, millions of pairs of portraits both with and without the extra light are used. Photographing such a dataset in real life would be challenging, requiring near-perfect registration of portraits captured across different lighting conditions. Instead, many individuals with different face shapes, genders, skin tones, hairstyles, and clothing/accessories can be photographed in a computational illumination system.
(73) Surface Orientation Map*Optimal Light Direction=Light Visibility Map
(74) FIG. **8**A illustrates an example network architecture to determine a light visibility map, in accordance with example embodiments. Input image **210** can be processed to output surface orientation map **220** and light direction **230**. As per Lambert's law, an object composed of a relatively matte material reflects an amount of light proportional to the cosine of the angle between its surface orientation (e.g., surface orientation map **220**) and the direction of the incident light (e.g., light direction **230**). Accordingly, Lambert's law can be applied to take the dot product **235** to obtain light visibility map **240**.
(75) Light visibility map **240** is a representation of light direction **230** customized to surface geometry. Such a representation can serve two purposes. First, light direction **230** is more closely tied to a prediction of surface geometry, which may greatly simplify the image quality assessment of relighting network **200**. Many errors made in relighting can be traced back to errors in the predicted surface orientation map **220**. Second, light visibility map **240** serves as a better per-pixel representation of light direction **230** for the quotient image prediction UNet, such as third convolution network **245** of FIG. **2**. Relighting network **200** need not infer which pixels are to be brightened, and only has to determine how they are brightened (e.g., material properties), and how subtle shadows may be cast (e.g., occlusions).
(76) FIG. **8**B illustrates an example image highlighting an effect of surface geometry in photo illumination, in accordance with example embodiments. For example, image **810** is obtained in a lighting network where a surface orientation map **220** is not used, and image **820** is obtained in a lighting network where a surface orientation map **220** is used. As illustrated, the geometry of the face in image **820** is more detailed and there is less flattening in the lower left half of the face, from the nose to the chin.
(77) Quotient Image—Learning Detail-Preserving Relighting
(78) FIG. **9** is a diagram depicting an example network to predict a quotient image, in accordance with example embodiments. For example, a convolutional neural network **245** may take input image **210**, and light visibility map **240** to predict a quotient image **250** indicative of a per pixel light energy needed to enhance input image **210**. Generally, quotient image **250** is a quotient of relit image **260** over input image **210**. Predicting such a multiplicative factor enables relighting network **200** to preserve high frequency details like skin pores and individual hair strands in input image **210**, rather than attempting to reproduce these high frequency details through a neural network. The predicted quotient image **250** is of significantly lower resolution than the input image **210**. In order to apply a lower resolution quotient image **250** to a higher resolution input image **210**, the predicted quotient image **250** can be upsampled using bilinear or other upsampling methods.
(79) FIG. **10** is an example architecture for a neural network to predict a quotient image, in accordance with example embodiments. For example, convolutional neural network **245** can be modeled around a UNet with skip connections. In some embodiments, the input image **210** and light visibility map **240** can go through 6 encoder blocks, Enc L1 **1010**, Enc L2 **1012**, Enc L3 **1014**, Enc L4 **1016**, Enc L5 **1018**, and Enc L6 **1020**, and 6 decoder blocks, Dec L1 **1022**, Dec L2 **1024**, Dec L3 **1026**, Dec L4 **1028**, Dec L5 **1030**, and Dec L6 **1032**. Each encoder block of encoder **1001** consists of a single convolution followed by downsampling by a factor of 2 followed by a blur-pooling operation. Each decoder block of decoder **1003** consists of a single convolution followed by upsampling by a factor of 2 followed by a bilinear upsampling operation. Skip connections from the encoder **1001** are concatenated or added to the upsampled features of decoder **1003** depending on the network version. In some example embodiments, filters for encoder **1001** may include (16,

32, 64, 128, 256, 256) filters for the encoder blocks yielding a final bottleneck size of (4×3×256). Similarly, filters for decoder **1003** may include (256, 256, 128, 64, 32, 16) features to return back to the (256×192) image resolution. A final convolution with 3 filters can be used to produce quotient image **250** from the decoder output. Relu activations can be used after all convolutions. This may be further supervised with the ground truth image data **1005**.

(80) Convolutional neural network **245** can be a fully-convolutional neural network as described herein. During training, convolutional neural network **245** can receive as inputs one or more input training images **1006** (**1** . . . M) and light data **1008** (**1** . . . M) from ground truth image data **1005**. Convolutional neural network **245** can include layers of nodes for processing input image **210** and light visibility map **240**. Example layers can include, but are not limited to, input layers, convolutional layers, activation layers, pooling layers, and output layers. Input layers can store input data, such as pixel data of input image **210** and light visibility map **240**, and inputs from other layers of convolutional neural network **245**. Convolutional layers can compute an output of neurons connected to local regions in the input. In some cases, a bilinear upsampling followed by a convolution is performed to apply a filter to a relatively small input to expand/upsample the relatively small input to become a larger output. Activation layers can determine whether or not an output of a preceding layer is “activated” or actually provided (e.g., provided to a succeeding layer). Pooling layers can downsample the input. For example, convolutional neural network **245** can use one or more pooling layers to downsample the input by a predetermined factor (e.g., a factor of two) in the horizontal and/or vertical dimensions. Output layers can provide an output of conventional neural network **245** to software and/or hardware interfacing with conventional neural network **245**; e.g. to hardware and/or software used to display, print, communicate and/or otherwise provide quotient image **250** (e.g., to one or more components of relighting network **200**). Layers **1010**, **1012**, **1014**, **1016**, **1018**, **1020**, **1022**, **1024**, **1026**, **1028**, **1030**, **1032** can include one or more input layers, output layers, convolutional layers, activation layers, pooling layers, and/or other layers described herein.

(81) In some implementations, convolutional neural network **245** can include encoding layers **1010**, **1012**, **1014**, **1016**, **1018** arranged respectively as in an order as layers L1, L2, L3, L4, L5 each successively convolving its input and providing its output to a successive layer until reaching encoding layer L6 **1020**. In FIG. **10**, a depicted layer can include one or more actual layers. For example, encoding layer L1 **1010** can have one or more input layers, one or more activation layers, and/or one or more additional layers. As another example, encoding layer L2 **1012**, encoding layer L3 **1014**, encoding layer L4 **1016**, encoding layer L5 **1018**, and/or encoding layer L6 **1020** can include one or more convolutional layers, one or more activation layers (e.g., having a one-to-one relationship to the one or more convolutional layer), one or more pooling layers, and/or one or more additional layers.

(82) In some examples, some or all of the pooling layers in convolutional neural network **245** can downsample an input by a common factor in both horizontal and vertical dimensions, while not downsampling depth dimensions associated with the input. The depth dimensions could store data for pixel colors (red, green, blue) and/or data representing scores. Other common factors for downsampling other than two can be used as well by one or more (pooling) layers of convolutional neural network **245**.

(83) Encoding layer L1 **1010** can receive and process input image **210** and light visibility map **240** and provide an output to encoding layer L2 **1012**. Encoding layer L2 **1012** can process the output of encoding layer L1 **1010** and provide an output to encoding layer L3 **1014**. Encoding layer L3 **1014** can process the output of encoding layer L2 **1012** and provide an output to encoding layer L4 **1016**. Encoding layer L4 **1016** can process the output of encoding layer L3 **1014** and provide an output to encoding layer L5 **1018**. Encoding layer L5 **1018** can process the output of encoding layer L4 **1016** and provide an output to encoding layer L6 **1020**.

(84) Encoding layer L6 **1020** may provide the output to decoding layer L1 **1022** to begin predicting

quotient image **250**. Decoding layer L2 **1024** can receive and process inputs from both decoding layer L1 **1022** and encoding layer L5 **1018** (e.g., using a skip connection between encoding layer L5 **1018** and decoding layer L2 **1024**) to provide an output to decoding layer L3 **1026**. Decoding layer L3 **1026** can receive and process inputs from both decoding layer L2 **1024** and encoding layer L4 **1016** (e.g., using a skip connection between encoding layer L4 **1016** and decoding layer L3 **1026**) to provide an output to decoding layer L4 **1028**. Decoding layer L4 **1028** can receive and process inputs from both decoding layer L3 **1026** and encoding layer L3 **1014** (e.g., using a skip connection between encoding layer L3 **1014** and decoding layer L4 **1028**) to provide an output to decoding layer L5 **1030**. Decoding layer L5 **1030** can receive and process inputs from both decoding layer L4 **1028** and encoding layer L2 **1012** (e.g., using a skip connection between encoding layer L2 **1012** and decoding layer L5 **1030**) to provide an output to decoding layer L6 **1032**. Decoding layer L6 **1032** can receive and process inputs from both decoding layer L5 **1030** and encoding layer L1 **1010** (e.g., using a skip connection between encoding layer L1 **1010** and decoding layer L6 **1032**) to provide a prediction of quotient image **250**, which can then be output from decoding layer L6 **1032**. The data provided by skip connections between encoding layers **1018**, **1016**, **1014**, **1012**, **1010** and respective decoding layers **1024**, **1026**, **1028**, **1030**, **1032** can be used by each respective decoding layer to provide additional details for generating a decoding layer's contribution to the prediction of quotient image **250**. In some examples, each of decoding layers **1022**, **1024**, **1026**, **1028**, **1030**, **1032** used to predict quotient image **250** can include one or more convolution layers, one or more activation layers, and perhaps one or more input and/or output layers. In some examples, some or all of layers **1010**, **1012**, **1014**, **1016**, **1018**, **1020**, **1022**, **1024**, **1026**, **1028**, **1030**, **1032** can act as a convolutional encoder/decoder network.

(85) In some implementations, convolutional neural network **245** can be trained end to end with losses on the high resolution relit image **260**. For example, a loss can be computed as Loss (UpsampledQuotient*HighResInputImage, HighResGroundTruthImage), where UnsampledQuotient is an unsampled quotient image, HighResInputImage is the high resolution input image, and HighResGroundTruthImage is the high resolution ground truth image, to directly focus on the high resolution relit image **260**, instead of attempting to obtain an exact quotient image **250**. A combination of L1 and adversarial losses can be used to train convolutional neural network **245**. Generally, empirical data suggests that adversarial loss is a significant factor for good generalization to images in the wild when training data is limited (e.g., 15 subjects). However, L1 loss achieved similar results with a larger dataset (e.g., 70 subjects). Furthermore, adversarial loss can become harder to train with a larger variation in viewpoints and subject clothing. In some implementations, adversarial loss may be selectively applied to the face portion of an image. Other loss measures can be used as well or instead. For example, an L2 loss measure between surface orientation map predictions and training images can be minimized during training of convolutional neural network **245** to obtain relit image **260**.

(86) As described herein, convolutional neural network **245** can include perceptual loss processing. For example, convolutional neural network **245** can use generative adversarial net (GAN) loss functions to determine if part or all of an image would be predicted to generate a surface orientation map, and so satisfy one or more perceptually-related conditions on lighting of that part of the image. In some examples, the predicted outputs can be fed back into the convolutional neural network **245** again as input to perform iterative refinement. In some examples, convolutional neural network **245** can utilize deep supervision techniques to provide constraints on intermediate layers. In some examples, convolutional neural network **245** can have more, fewer, and/or different layers to those shown in FIG. **10**.

(87) FIG. **11** is an example network illustrating post-processing techniques, in accordance with example embodiments. In one aspect, a multiplication/post-processing operation **1105** can be applied to input image **210** and quotient image **250** to output relit image **260**. As described herein, quotient image **250** is indicative of a per pixel multiplicative factor that makes each pixel darker,

lighter, or unchanged. Values of quotient image **250** are real numbers that are not constrained. However, during post-processing **1105**, an increase in average foreground brightness, or average foreground luma, can be bounded. Such an increase can be kept fixed during training, and during post-processing **1105**, constraints may be added so that the relit image **260** is photorealistic. Accordingly, in some embodiments, quotient image **250** may be post-processed, thereby resulting in a more realistic rendition of relit image **260**.

(88) One or more post-processing techniques can be applied. For example, highlight protection **1110** can be applied to enhance input image **210**. Brightly lit portions of the input image **210** can be subdued, thereby providing a more realistic rendition of input image **210** by restoring, for example, a natural skin color. Generally, highlight protection **1110** provides a local adjustment of brightness. Also, for example, exposure compensation **1115** can be applied. Assuming that input image **210** is captured with a fixed exposure, the input image **210** can be normalized to a fixed foreground exposure value. Generally, input images may have a wide range of exposure values. For example, exposure values can depend on light colored clothes, dark colored clothes, and so on. Thus, input image **210** may be compensated for under and/or over exposure, and exposure can be compensated to further adjust the quotient images. Generally, exposure compensation **1115** provides a global adjustment of brightness. As another example, a matting refinement **1120**, can be applied to input image **210** to smooth out edges.

(89) FIG. **12** is an example interactive graphical user interface (UI) **1200**, in accordance with example embodiments. UI **1200** can be displayed via a computing device, such as a display screen **1210** of a mobile phone. A user can be provided with an ability to adjust a light position **1220** by changing input values via a user-friendly slider bar **1230**. For example, a selectable option for "Portrait Light" **1240** can be selected by the user. This allows the user to reposition the light position **1220** to relight an image. The user can also change light intensity by using a user-friendly intensity scale **1250**. As another example, an "Auto" setting **1260** can automatically reposition light position **1220** and light intensity for optimal effect. Adjusting the light position and light intensity changes the synthetic illumination, and an input image can be enhanced based on the one or more techniques described herein (e.g., with reference to relighting network **200**).

(90) FIG. **13** illustrates an example input-output image pair based on a surface orientation map, in accordance with example embodiments. For example, input image **1310** can be processed to obtain surface orientation map **1320**. The surface orientation map **1320** can then be combined with a light direction to output relit image **1330**.

(91) FIG. **14** illustrates an example relighting pipeline, in accordance with example embodiments. Input image **1410** shows a portrait of a human. In some instances, input image **1410** can be an image that a user has captured with an image capturing device. In other instances, input image **1410** can be an image from a photo library of a user. A user may provide an instruction to enhance input image **1410**. In response, input image **1410** is processed for light estimation **1420**. For example, an original environmental lighting in input image **1410** can be predicted. In some implementations, a convolutional neural network (e.g., convolutional neural network **225**) can be used for light estimation **1420**. A face detection operation can be performed on input image **1420**, and a face light estimation **1430** can be performed. In some instances, one or more aspects of the network architecture described with respect to FIG. **6** can be used for face light estimation **1430**.

(92) FIG. **15** illustrates example input images with predicted face light estimations, in accordance with example embodiments. A face light estimation **1530** can be performed on each input image. Each input image in FIG. **15** is shown with two sets of 3 spheres. The first set (to the left) corresponds to estimated face lighting, and the second set (to the right) corresponds to an environmental lighting or ground truth. The three spheres provide visual representations of color, intensity, and directionality of the environmental lighting or ground truth in each respective input image.

(93) Referring to FIG. **14** again, segmentation **1440** can be performed on input image **1410**. For

example, a body part segmentation or an alpha mask segmentation can be performed to detect foreground object **1450**. Input image **1410**, predicted face lighting **1430**, and foreground object **1450** can be input to relighting network **1460**. One or more aspects of relighting network **1460** can correspond to aspects of relighting network **200**. For example, a light direction for synthetic lighting can be automatically optimized. In some instances, aspects of this operation can be performed by a neural network (e.g., convolutional neural network **225**). In some implementations, the light direction can be input as a user preference (e.g., via user interface **1200**). Also, for example, a surface orientation map (e.g., surface orientation map **220**) can be predicted. In some instances, aspects of this operation can be performed by a neural network (e.g., convolutional neural network **215**). A scalar product of the surface orientation map and the light direction can be used to output a light visibility map (e.g., light visibility map **240**). A quotient image **1470** can be output by relighting network **1460**. In some instances, input image **1410** and quotient image **1470** can be multiplied to output relit image **1490**. In some instances, post-processing **1480** can be performed on input image **1410** based on quotient image **1470** to output relit image **1490**.

(94) Training Machine Learning Models for Generating Inferences/Predictions

(95) FIG. **16** shows diagram **1600** illustrating a training phase **1602** and an inference phase **1604** of trained machine learning model(s) **1632**, in accordance with example embodiments. Some machine learning techniques involve training one or more machine learning algorithms, on an input set of training data to recognize patterns in the training data and provide output inferences and/or predictions about (patterns in the) training data. The resulting trained machine learning algorithm can be termed as a trained machine learning model. For example, FIG. **16** shows training phase **1602** where one or more machine learning algorithms **1620** are being trained on training data **1610** to become trained machine learning model **1632**. Then, during inference phase **1604**, trained machine learning model **1632** can receive input data **1630** and one or more inference/prediction requests **1640** (perhaps as part of input data **1630**) and responsively provide as an output one or more inferences and/or predictions **1650**.

(96) As such, trained machine learning model(s) **1632** can include one or more models of one or more machine learning algorithms **1620**. Machine learning algorithm(s) **1620** may include, but are not limited to: an artificial neural network (e.g., a herein-described convolutional neural networks, a recurrent neural network, a Bayesian network, a hidden Markov model, a Markov decision process, a logistic regression function, a support vector machine, a suitable statistical machine learning algorithm, and/or a heuristic machine learning system). Machine learning algorithm(s) **1620** may be supervised or unsupervised, and may implement any suitable combination of online and offline learning.

(97) In some examples, machine learning algorithm(s) **1620** and/or trained machine learning model(s) **1632** can be accelerated using on-device coprocessors, such as graphic processing units (GPUs), tensor processing units (TPUs), digital signal processors (DSPs), and/or application specific integrated circuits (ASICs). Such on-device coprocessors can be used to speed up machine learning algorithm(s) **1620** and/or trained machine learning model(s) **1632**. In some examples, trained machine learning model(s) **1632** can be trained, reside and execute to provide inferences on a particular computing device, and/or otherwise can make inferences for the particular computing device.

(98) During training phase **1602**, machine learning algorithm(s) **1620** can be trained by providing at least training data **1610** as training input using unsupervised, supervised, semi-supervised, and/or reinforcement learning techniques. Unsupervised learning involves providing a portion (or all) of training data **1610** to machine learning algorithm(s) **1620** and machine learning algorithm(s) **1620** determining one or more output inferences based on the provided portion (or all) of training data **1610**. Supervised learning involves providing a portion of training data **1610** to machine learning algorithm(s) **1620**, with machine learning algorithm(s) **1620** determining one or more output inferences based on the provided portion of training data **1610**, and the output inference(s) are

either accepted or corrected based on correct results associated with training data **1610**. In some examples, supervised learning of machine learning algorithm(s) **1620** can be governed by a set of rules and/or a set of labels for the training input, and the set of rules and/or set of labels may be used to correct inferences of machine learning algorithm(s) **1620**.

(99) Semi-supervised learning involves having correct results for part, but not all, of training data **1610**. During semi-supervised learning, supervised learning is used for a portion of training data **1610** having correct results, and unsupervised learning is used for a portion of training data **1610** not having correct results. Reinforcement learning involves machine learning algorithm(s) **1620** receiving a reward signal regarding a prior inference, where the reward signal can be a numerical value. During reinforcement learning, machine learning algorithm(s) **1620** can output an inference and receive a reward signal in response, where machine learning algorithm(s) **1620** are configured to try to maximize the numerical value of the reward signal. In some examples, reinforcement learning also utilizes a value function that provides a numerical value representing an expected total of the numerical values provided by the reward signal over time. In some examples, machine learning algorithm(s) **1620** and/or trained machine learning model(s) **1632** can be trained using other machine learning techniques, including but not limited to, incremental learning and curriculum learning.

(100) In some examples, machine learning algorithm(s) **1620** and/or trained machine learning model(s) **1632** can use transfer learning techniques. For example, transfer learning techniques can involve trained machine learning model(s) **1632** being pre-trained on one set of data and additionally trained using training data **1610**. More particularly, machine learning algorithm(s) **1620** can be pre-trained on data from one or more computing devices and a resulting trained machine learning model provided to a particular computing device, where the particular computing device is intended to execute the trained machine learning model during inference phase **1604**. Then, during training phase **1602**, the pre-trained machine learning model can be additionally trained using training data **1610**, where training data **1610** can be derived from kernel and non-kernel data of the particular computing device. This further training of the machine learning algorithm(s) **1620** and/or the pre-trained machine learning model using training data **1610** of the particular computing device's data can be performed using either supervised or unsupervised learning. Once machine learning algorithm(s) **1620** and/or the pre-trained machine learning model has been trained on at least training data **1610**, training phase **1602** can be completed. The trained resulting machine learning model can be utilized as at least one of trained machine learning model(s) **1632**.

(101) In particular, once training phase **1602** has been completed, trained machine learning model(s) **1632** can be provided to a computing device, if not already on the computing device. Inference phase **1604** can begin after trained machine learning model(s) **1632** are provided to the particular computing device.

(102) During inference phase **1604**, trained machine learning model(s) **1632** can receive input data **1630** and generate and output one or more corresponding inferences and/or predictions **1650** about input data **1630**. As such, input data **1630** can be used as an input to trained machine learning model(s) **1632** for providing corresponding inference(s) and/or prediction(s) **1650** to kernel components and non-kernel components. For example, trained machine learning model(s) **1632** can generate inference(s) and/or prediction(s) **1650** in response to one or more inference/prediction requests **1640**. In some examples, trained machine learning model(s) **1632** can be executed by a portion of other software. For example, trained machine learning model(s) **1632** can be executed by an inference or prediction daemon to be readily available to provide inferences and/or predictions upon request. Input data **1630** can include data from the particular computing device executing trained machine learning model(s) **1632** and/or input data from one or more computing devices other than the particular computing device.

(103) Input data **1630** can include a collection of images provided by one or more sources. The

collection of images can include images of an object, such as a human face, where the images of the human face are taken under different lighting conditions, images of multiple objects, images resident on the particular computing device, and/or other images. Other types of input data are possible as well.

(104) Inference(s) and/or prediction(s) **1650** can include output images, output lighting models, output surface orientation maps, output light estimations, output quotient images, numerical values, and/or other output data produced by trained machine learning model(s) **1632** operating on input data **1630** (and training data **1610**). In some examples, trained machine learning model(s) **1632** can use output inference(s) and/or prediction(s) **1650** as input feedback **1660**. Trained machine learning model(s) **1632** can also rely on past inferences as inputs for generating new inferences.

(105) Convolutional neural networks **215**, **225**, **245** can be examples of machine learning algorithm(s) **1620**. After training, the trained version of convolutional neural networks **215**, **225**, **245** can be examples of trained machine learning model(s) **1632**. In this approach, an example of inference/prediction request(s) **1640** can be a request to predict a surface orientation map from an input image of an object and a corresponding example of inferences and/or prediction(s) **1650** can be an output surface orientation map. As another example, an example of inference/prediction request(s) **1640** can be a request to predict an environmental lighting from an input image of an object and a corresponding example of inferences and/or prediction(s) **1650** can be an output set of 3-D vectors that predicts application of a particular lighting direction to the input image. Also, for example, an example of inference/prediction request(s) **1640** can be a request to determine a quotient image for an input image of an object and a corresponding example of inferences and/or prediction(s) **1650** can be an output image that predicts the quotient image as a per pixel multiplicative factor to be applied to the input image.

(106) In some examples, a single computing device ("CD_SOLO") can include the trained version of convolutional neural network **215**, perhaps after training convolutional neural network **215**. Then, computing device CD_SOLO can receive requests to predict surface orientation maps from corresponding input images, and use the trained version of convolutional neural network **215** to generate output images that predict the surface orientation maps. In some examples, one computing device CD_SOLO can include the trained version of convolutional neural network **225**, perhaps after training convolutional neural network **225**. Then, computing device CD_SOLO can receive requests to predict light directions from corresponding input images, and use the trained version of convolutional neural network **225** to generate output images that predict the light directions. In some examples, one computing device CD_SOLO can include the trained version of convolutional neural network **245**, perhaps after training convolutional neural network **245**. Then, computing device CD_SOLO can receive requests to predict quotient images from corresponding input images, and use the trained version of convolutional neural network **215** to generate output images that predict the quotient images.

(107) In some examples, two or more computing devices, such as a first client device ("CD_CLI") and a server device ("CD_SRV") can be used to provide output images; e.g., a first computing device CD_CLI can generate and send requests to apply particular synthetic lighting to corresponding input images to a second computing device CD_SRV. Then, CD_SRV can use the trained versions of convolutional neural networks **215**, **225**, **245**, perhaps after training convolutional neural networks **215**, **225**, **245**, to generate output images that predict application of the particular synthetic lighting to the input images, and respond to the requests from CD_CLI for the output images. Then, upon reception of responses to the requests, CD_CLI can provide the requested output images (e.g., using a user interface and/or a display, a printed copy, an electronic communication, etc.).

(108) Example Data Network

(109) FIG. **17** depicts a distributed computing architecture **1700**, in accordance with example embodiments. Distributed computing architecture **1700** includes server devices **1708**, **1710** that are

configured to communicate, via network **1706**, with programmable devices **1704***a*, **1704***b*, **1704***c*, **1704***d*, **1704***e*. Network **1706** may correspond to a local area network (LAN), a wide area network (WAN), a WLAN, a WWAN, a corporate intranet, the public Internet, or any other type of network configured to provide a communications path between networked computing devices. Network **1706** may also correspond to a combination of one or more LANs, WANs, corporate intranets, and/or the public Internet.

(110) Although FIG. **17** only shows five programmable devices, distributed application architectures may serve tens, hundreds, or thousands of programmable devices. Moreover, programmable devices **1704***a*, **1704***b*, **1704***c*, **1704***d*, **1704***e* (or any additional programmable devices) may be any sort of computing device, such as a mobile computing device, desktop computer, wearable computing device, head-mountable device (HMD), network terminal, a mobile computing device, and so on. In some examples, such as illustrated by programmable devices **1704***a*, **1704***b*, **1704***c*, **1704***e*, programmable devices can be directly connected to network **1706**. In other examples, such as illustrated by programmable device **1704***d*, programmable devices can be indirectly connected to network **1706** via an associated computing device, such as programmable device **1704***c*. In this example, programmable device **1704***c* can act as an associated computing device to pass electronic communications between programmable device **1704***d* and network **1706**. In other examples, such as illustrated by programmable device **1704***e*, a computing device can be part of and/or inside a vehicle, such as a car, a truck, a bus, a boat or ship, an airplane, etc. In other examples not shown in FIG. **17**, a programmable device can be both directly and indirectly connected to network **1706**.

(111) Server devices **1708**, **1710** can be configured to perform one or more services, as requested by programmable devices **1704***a*-**1704***e*. For example, server device **1708** and/or **1710** can provide content to programmable devices **1704***a*-**1704***e*. The content can include, but is not limited to, web pages, hypertext, scripts, binary data such as compiled software, images, audio, and/or video. The content can include compressed and/or uncompressed content. The content can be encrypted and/or unencrypted. Other types of content are possible as well.

(112) As another example, server device **1708** and/or **1710** can provide programmable devices **1704***a*-**1704***e* with access to software for database, search, computation, graphical, audio, video, World Wide Web/Internet utilization, and/or other functions. Many other examples of server devices are possible as well.

(113) Computing Device Architecture

(114) FIG. **18** is a block diagram of an example computing device **1800**, in accordance with example embodiments. In particular, computing device **1800** shown in FIG. **18** can be configured to perform at least one function of and/or related to a convolutional neural network, a predicted surface orientation map, a predicted original lighting model, a predicted quotient image, convolutional neural networks **215**, **225**, **245**, and/or method **2000**.

(115) Computing device **1800** may include a user interface module **1801**, a network communications module **1802**, one or more processors **1803**, data storage **1804**, one or more cameras **1818**, one or more sensors **1820**, and power system **1822**, all of which may be linked together via a system bus, network, or other connection mechanism **1805**.

(116) User interface module **1801** can be operable to send data to and/or receive data from external user input/output devices. For example, user interface module **1801** can be configured to send and/or receive data to and/or from user input devices such as a touch screen, a computer mouse, a keyboard, a keypad, a touch pad, a trackball, a joystick, a voice recognition module, and/or other similar devices. User interface module **1801** can also be configured to provide output to user display devices, such as one or more cathode ray tubes (CRT), liquid crystal displays, light emitting diodes (LEDs), displays using digital light processing (DLP) technology, printers, light bulbs, and/or other similar devices, either now known or later developed. User interface module **1801** can also be configured to generate audible outputs, with devices such as a speaker, speaker jack, audio

output port, audio output device, earphones, and/or other similar devices. User interface module **1801** can further be configured with one or more haptic devices that can generate haptic outputs, such as vibrations and/or other outputs detectable by touch and/or physical contact with computing device **1800**. In some examples, user interface module **1801** can be used to provide a graphical user interface (GUI) for utilizing computing device **1800**, such as, for example, a graphical user interface illustrated in FIG. **15**.

(117) Network communications module **1802** can include one or more devices that provide one or more wireless interfaces **1807** and/or one or more wireline interfaces **1808** that are configurable to communicate via a network. Wireless interface(s) **1807** can include one or more wireless transmitters, receivers, and/or transceivers, such as a Bluetooth™ transceiver, a Zigbee® transceiver, a Wi-Fi™ transceiver, a WiMAX™ transceiver, an LTE™ transceiver, and/or other type of wireless transceiver configurable to communicate via a wireless network. Wireline interface(s) **1808** can include one or more wireline transmitters, receivers, and/or transceivers, such as an Ethernet transceiver, a Universal Serial Bus (USB) transceiver, or similar transceiver configurable to communicate via a twisted pair wire, a coaxial cable, a fiber-optic link, or a similar physical connection to a wireline network.

(118) In some examples, network communications module **1802** can be configured to provide reliable, secured, and/or authenticated communications. For each communication described herein, information for facilitating reliable communications (e.g., guaranteed message delivery) can be provided, perhaps as part of a message header and/or footer (e.g., packet/message sequencing information, encapsulation headers and/or footers, size/time information, and transmission verification information such as cyclic redundancy check (CRC) and/or parity check values). Communications can be made secure (e.g., be encoded or encrypted) and/or decrypted/decoded using one or more cryptographic protocols and/or algorithms, such as, but not limited to, Data Encryption Standard (DES), Advanced Encryption Standard (AES), a Rivest-Shamir-Adelman (RSA) algorithm, a Diffie-Hellman algorithm, a secure sockets protocol such as Secure Sockets Layer (SSL) or Transport Layer Security (TLS), and/or Digital Signature Algorithm (DSA). Other cryptographic protocols and/or algorithms can be used as well or in addition to those listed herein to secure (and then decrypt/decode) communications.

(119) One or more processors **1803** can include one or more general purpose processors, and/or one or more special purpose processors (e.g., digital signal processors, tensor processing units (TPUs), graphics processing units (GPUs), application specific integrated circuits, etc.). One or more processors **1803** can be configured to execute computer-readable instructions **1806** that are contained in data storage **1804** and/or other instructions as described herein.

(120) Data storage **1804** can include one or more non-transitory computer-readable storage media that can be read and/or accessed by at least one of one or more processors **1803**. The one or more computer-readable storage media can include volatile and/or non-volatile storage components, such as optical, magnetic, organic or other memory or disc storage, which can be integrated in whole or in part with at least one of one or more processors **1803**. In some examples, data storage **1804** can be implemented using a single physical device (e.g., one optical, magnetic, organic or other memory or disc storage unit), while in other examples, data storage **1804** can be implemented using two or more physical devices.

(121) Data storage **1804** can include computer-readable instructions **1806** and perhaps additional data. In some examples, data storage **1804** can include storage required to perform at least part of the herein-described methods, scenarios, and techniques and/or at least part of the functionality of the herein-described devices and networks. In some examples, data storage **1804** can include storage for a trained neural network model **1812** (e.g., a model of trained convolutional neural networks such as convolutional neural networks **215**, **225**, **245**). In particular of these examples, computer-readable instructions **1806** can include instructions that, when executed by processor(s) **1803**, enable computing device **1800** to provide for some or all of the functionality of trained

neural network model **1812**.

(122) In some examples, computing device **1800** can include one or more cameras **1818**. Camera(s) **1818** can include one or more image capture devices, such as still and/or video cameras, equipped to capture light and record the captured light in one or more images; that is, camera(s) **1818** can generate image(s) of captured light. The one or more images can be one or more still images and/or one or more images utilized in video imagery. Camera(s) **1818** can capture light and/or electromagnetic radiation emitted as visible light, infrared radiation, ultraviolet light, and/or as one or more other frequencies of light.

(123) In some examples, computing device **1800** can include one or more sensors **1820**. Sensors **1820** can be configured to measure conditions within computing device **1800** and/or conditions in an environment of computing device **1800** and provide data about these conditions. For example, sensors **1820** can include one or more of: (i) sensors for obtaining data about computing device **1800**, such as, but not limited to, a thermometer for measuring a temperature of computing device **1800**, a battery sensor for measuring power of one or more batteries of power system **1822**, and/or other sensors measuring conditions of computing device **1800**; (ii) an identification sensor to identify other objects and/or devices, such as, but not limited to, a Radio Frequency Identification (RFID) reader, proximity sensor, one-dimensional barcode reader, two-dimensional barcode (e.g., Quick Response (QR) code) reader, and a laser tracker, where the identification sensors can be configured to read identifiers, such as RFID tags, barcodes, QR codes, and/or other devices and/or object configured to be read and provide at least identifying information; (iii) sensors to measure locations and/or movements of computing device **1800**, such as, but not limited to, a tilt sensor, a gyroscope, an accelerometer, a Doppler sensor, a GPS device, a sonar sensor, a radar device, a laser-displacement sensor, and a compass; (iv) an environmental sensor to obtain data indicative of an environment of computing device **1800**, such as, but not limited to, an infrared sensor, an optical sensor, a light sensor, a biosensor, a capacitive sensor, a touch sensor, a temperature sensor, a wireless sensor, a radio sensor, a movement sensor, a microphone, a sound sensor, an ultrasound sensor and/or a smoke sensor; and/or (v) a force sensor to measure one or more forces (e.g., inertial forces and/or G-forces) acting about computing device **1800**, such as, but not limited to one or more sensors that measure: forces in one or more dimensions, torque, ground force, friction, and/or a zero moment point (ZMP) sensor that identifies ZMPs and/or locations of the ZMPs. Many other examples of sensors **1820** are possible as well.

(124) Power system **1822** can include one or more batteries **1824** and/or one or more external power interfaces **1826** for providing electrical power to computing device **1800**. Each battery of the one or more batteries **1824** can, when electrically coupled to the computing device **1800**, act as a source of stored electrical power for computing device **1800**. One or more batteries **1824** of power system **1822** can be configured to be portable. Some or all of one or more batteries **1824** can be readily removable from computing device **1800**. In other examples, some or all of one or more batteries **1824** can be internal to computing device **1800**, and so may not be readily removable from computing device **1800**. Some or all of one or more batteries **1824** can be rechargeable. For example, a rechargeable battery can be recharged via a wired connection between the battery and another power supply, such as by one or more power supplies that are external to computing device **1800** and connected to computing device **1800** via the one or more external power interfaces. In other examples, some or all of one or more batteries **1824** can be non-rechargeable batteries.

(125) One or more external power interfaces **1826** of power system **1822** can include one or more wired-power interfaces, such as a USB cable and/or a power cord, that enable wired electrical power connections to one or more power supplies that are external to computing device **1800**. One or more external power interfaces **1826** can include one or more wireless power interfaces, such as a Qi wireless charger, that enable wireless electrical power connections, such as via a Qi wireless charger, to one or more external power supplies. Once an electrical power connection is established to an external power source using one or more external power interfaces **1826**, computing device

1800 can draw electrical power from the external power source the established electrical power connection. In some examples, power system **1822** can include related sensors, such as battery sensors associated with the one or more batteries or other types of electrical power sensors.
(126) Cloud-Based Servers
(127) FIG. **19** depicts a network **1706** of computing clusters **1909***a*, **1909***b*, **1909***c* arranged as a cloud-based server system in accordance with an example embodiment. Computing clusters **1909***a*, **1909***b*, **1909***c* can be cloud-based devices that store program logic and/or data of cloud-based applications and/or services; e.g., perform at least one function of and/or related to a convolutional neural networks, a predicted surface orientation map, a predicted light direction, a predicted quotient image, convolutional neural networks **215**, **225**, **245**, and/or method **2400**.
(128) In some embodiments, computing clusters **1909***a*, **1909***b*, **1909***c* can be a single computing device residing in a single computing center. In other embodiments, computing clusters **1909***a*, **1909***b*, **1909***c* can include multiple computing devices in a single computing center, or even multiple computing devices located in multiple computing centers located in diverse geographic locations. For example, FIG. **19** depicts each of computing clusters **1909***a*, **1909***b*, and **1909***c* residing in different physical locations.
(129) In some embodiments, data and services at computing clusters **1909***a*, **1909***b*, **1909***c* can be encoded as computer readable information stored in non-transitory, tangible computer readable media (or computer readable storage media) and accessible by other computing devices. In some embodiments, computing clusters **1909***a*, **1909***b*, **1909***c* can be stored on a single disk drive or other tangible storage media, or can be implemented on multiple disk drives or other tangible storage media located at one or more diverse geographic locations.
(130) FIG. **19** depicts a cloud-based server system in accordance with an example embodiment. In FIG. **19**, functionality of convolutional neural networks **215**, **225**, **245**, and/or a computing device can be distributed among computing clusters **1909***a*, **1909***b*, **1909***c*. Computing cluster **1909***a* can include one or more computing devices **1900***a*, cluster storage arrays **1910***a*, and cluster routers **1911***a* connected by a local cluster network **1912***a*. Similarly, computing cluster **1909***b* can include one or more computing devices **1900***b*, cluster storage arrays **1910***b*, and cluster routers **1911***b* connected by a local cluster network **1912***b*. Likewise, computing cluster **1909***c* can include one or more computing devices **1900***c*, cluster storage arrays **1910***c*, and cluster routers **1911***c* connected by a local cluster network **1912***c*.
(131) In some embodiments, each of computing clusters **1909***a*, **1909***b*, and **1909***c* can have an equal number of computing devices, an equal number of cluster storage arrays, and an equal number of cluster routers. In other embodiments, however, each computing cluster can have different numbers of computing devices, different numbers of cluster storage arrays, and different numbers of cluster routers. The number of computing devices, cluster storage arrays, and cluster routers in each computing cluster can depend on the computing task or tasks assigned to each computing cluster.
(132) In computing cluster **1909***a*, for example, computing devices **1900***a* can be configured to perform various computing tasks of convolutional neural network, confidence learning, and/or a computing device. In one embodiment, the various functionalities of a convolutional neural network, confidence learning, and/or a computing device can be distributed among one or more of computing devices **1900***a*, **1900***b*, **1900***c*. Computing devices **1900***b* and **1900***c* in respective computing clusters **1909***b* and **1909***c* can be configured similarly to computing devices **1900***a* in computing cluster **1909***a*. On the other hand, in some embodiments, computing devices **1900***a*, **1900***b*, and **1900***c* can be configured to perform different functions.
(133) In some embodiments, computing tasks and stored data associated with a convolutional neural networks, and/or a computing device can be distributed across computing devices **1900***a*, **1900***b*, and **1900***c* based at least in part on the processing requirements of a convolutional neural networks, and/or a computing device, the processing capabilities of computing devices **1900***a*,

1900*b*, **1900***c*, the latency of the network links between the computing devices in each computing cluster and between the computing clusters themselves, and/or other factors that can contribute to the cost, speed, fault-tolerance, resiliency, efficiency, and/or other design goals of the overall system architecture.

(134) Cluster storage arrays **1910***a*, **1910***b*, **1910***c* of computing clusters **1909***a*, **1909***b*, **1909***c* can be data storage arrays that include disk array controllers configured to manage read and write access to groups of hard disk drives. The disk array controllers, alone or in conjunction with their respective computing devices, can also be configured to manage backup or redundant copies of the data stored in the cluster storage arrays to protect against disk drive or other cluster storage array failures and/or network failures that prevent one or more computing devices from accessing one or more cluster storage arrays.

(135) Similar to the manner in which the functions of convolutional neural networks, and/or a computing device can be distributed across computing devices **1900***a*, **1900***b*, **1900***c* of computing clusters **1909***a*, **1909***b*, **1909***c*, various active portions and/or backup portions of these components can be distributed across cluster storage arrays **1910***a*, **1910***b*, **1910***c*. For example, some cluster storage arrays can be configured to store one portion of the data of a convolutional neural network, and/or a computing device, while other cluster storage arrays can store other portion(s) of data of a convolutional neural network, and/or a computing device. Also, for example, some cluster storage arrays can be configured to store the data of a first convolutional neural network, while other cluster storage arrays can store the data of a second and/or third convolutional neural network. Additionally, some cluster storage arrays can be configured to store backup versions of data stored in other cluster storage arrays.

(136) Cluster routers **1911***a*, **1911***b*, **1911***c* in computing clusters **1909***a*, **1909***b*, **1909***c* can include networking equipment configured to provide internal and external communications for the computing clusters. For example, cluster routers **1911***a* in computing cluster **1909***a* can include one or more internet switching and routing devices configured to provide (i) local area network communications between computing devices **1900***a* and cluster storage arrays **1910***a* via local cluster network **1912***a*, and (ii) wide area network communications between computing cluster **1909***a* and computing clusters **1909***b* and **1909***c* via wide area network link **1913***a* to network **1706**. Cluster routers **1911***b* and **1911***c* can include network equipment similar to cluster routers **1911***a*, and cluster routers **1911***b* and **1911***c* can perform similar networking functions for computing clusters **1909***b* and **1909***b* that cluster routers **1911***a* perform for computing cluster **1909***a*.

(137) In some embodiments, the configuration of cluster routers **1911***a*, **1911***b*, **1911***c* can be based at least in part on the data communication requirements of the computing devices and cluster storage arrays, the data communications capabilities of the network equipment in cluster routers **1911***a*, **1911***b*, **1911***c*, the latency and throughput of local cluster networks **1912***a*, **1912***b*, **1912***c*, the latency, throughput, and cost of wide area network links **1913***a*, **1913***b*, **1913***c*, and/or other factors that can contribute to the cost, speed, fault-tolerance, resiliency, efficiency and/or other design criteria of the moderation system architecture.

(138) Example Methods of Operation

(139) FIG. **20** is a flowchart of a method **2000**, in accordance with example embodiments. Method **2000** can be executed by a computing device, such as computing device **1700**. Method **2000** can begin at block **2010**, where the computing device can apply a geometry model to an input image to determine a surface orientation map indicative of a distribution of lighting on an object in the input image based on a surface geometry of the object, such as discussed above at least in the context of FIGS. **2-5**.

(140) At block **2020**, the computing device can apply an environmental light estimation model to the input image to determine a direction of synthetic lighting to be applied to the input image to enhance at least a portion of the input image, such as discussed above at least in the context of FIGS. **2**, **6**, and **7**.

(141) At block **2030**, the computing device can apply, based on the surface orientation map and the direction of synthetic lighting to be applied, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the input image, such as discussed above at least in the context of FIGS. **2**, **9-11**.
(142) At block **2040**, the computing device can enhance, based on the quotient image, the portion of the input image, such as discussed above at least in the context of FIGS. **2-15**.
(143) Some embodiments involve training a neural network to perform one or more of: (1) apply the geometry model to predict the surface orientation map, (2) apply the environmental light estimation model to determine the direction of the synthetic lighting, or (3) apply the light energy model to predict the quotient image. In some embodiments, the neural network is trained to apply the geometry model to predict the surface orientation map, and the embodiments may involve predicting the surface orientation map by using the trained neural network. In some embodiments, the neural network is trained to apply the environmental light estimation model to determine the direction of the synthetic lighting, and the embodiments may involve predicting the direction of the synthetic lighting by using the trained neural network. In some embodiments, the neural network is trained to apply the light energy model to predict the quotient image, and the embodiments may involve predicting the quotient by using the trained neural network.
(144) In some embodiments, the applying of the environmental light estimation model involves detecting, by the computing device, a pose of the object in the input image. The determining of the direction of the synthetic lighting can be based on the pose.
(145) Some embodiments involve generating, by the computing device and based on the surface orientation map and the direction of synthetic lighting, a light visibility map. The quotient image may be based on the light visibility map.
(146) In some embodiments, the enhancing of the portion of the input image involves determining, by the computing device, a request to enhance the portion of the input image. Such embodiments may also involve sending the request to enhance the portion of the input image from the computing device to a second computing device, the second computing device comprising the trained neural network. Such embodiments may also involve, after sending the request, the computing device receiving, from the second computing device, an output image that applies the quotient image to enhance the portion of the input image.
(147) In some embodiments, the training of the neural network involves utilizing a plurality of images of the object. The plurality of images may utilize a plurality of illumination profiles to light the object.
(148) Some embodiments involve receiving, by the computing device, a user preference for the direction of the synthetic lighting to be applied to the particular input image.
(149) In some embodiments, the object may have a characteristic of diffusely reflecting light.
(150) In some embodiments, the computing device may include a camera. Such embodiments involve generating the input image of the object using the camera. Such embodiments may also involve receiving, at the computing device, the generated input image from the camera.
(151) Some embodiments involve providing the portion of the enhanced input image using the computing device.
(152) Some embodiments involve predicting an illumination profile of the input image using the environmental light estimation model. Such embodiments involve providing the predicted illumination profile using the computing device.
(153) In some embodiments, the prediction of the illumination profile involves generating a high dynamic range (HDR) lighting environment based on low dynamic range (LDR) images of a set of reference objects, each of the set of reference objects having a respective bidirectional reflectance distribution function (BRDF). In such embodiments, the set of reference objects may include one or more of a mirror ball, a matte silver ball, or a gray diffuse ball.
(154) In some embodiments, the predicting involves obtaining the trained neural network at the

computing device. Such embodiments involve applying the obtained neural network at the computing device.

(155) In some embodiments, the training of the neural network includes training the neural network at the computing device.

(156) Some embodiments involve adjusting the quotient image to apply one or more of: (i) a compensation for an exposure level in the input image, (ii) a compensation for a brightness level in the input image, or (iii) a matting refinement to the particular input image.

Additional Example Embodiments

(157) The following clauses are offered as further description of the disclosure.

(158) Clause 1—A computer-implemented method, including: training a neural network to apply an environmental light estimation model to an input image to optimize a direction of synthetic lighting to be applied to the input image to enhance at least a portion of the input image; detecting, by a computing device, a pose of an object in a particular input image; and predicting, by the computing device and based on the pose, a direction of synthetic lighting to be applied to the input image by using the trained neural network to apply the predicted environmental light estimation model to the particular input image.

(159) Clause 2—The computer-implemented method of clause 1, further including: applying the optimized direction of synthetic lighting to the particular input image to enhance the portion of the particular input image.

(160) Clause 3—The computer-implemented method of clause 2, wherein applying the optimized direction of synthetic lighting to the particular input image includes: determining, by the computing device, a request to enhance the particular input image; sending the request to enhance the particular input image from the computing device to a second computing device, the second computing device including the trained neural network; and after sending the request, the computing device receiving, from the second computing device, an output image that applies the optimized direction of synthetic lighting to the particular input image.

(161) Clause 4—The computer-implemented method of any one of clauses 1-3, wherein an illumination profile of the input image is modeled using a light estimation model, and wherein predicting the optimal direction from which to add synthetic lighting further comprises estimating an original illumination profile for the input image.

(162) Clause 5—The computer-implemented method of any one of clauses 1-4, wherein the particular input image includes a portrait, and the method further including: detecting a face in the portrait; and determining the pose of the object by determining ahead pose of the face.

(163) Clause 6—The computer-implemented method of any one of clauses 1-5, wherein training the neural network to apply the environmental light estimation model comprises training the neural network using a plurality of images of the object, where the plurality of images utilize a plurality of illumination profiles to light the object.

(164) Clause 7—The computer-implemented method of any one of clauses 1-6, wherein the neural network is a convolutional neural network.

(165) Clause 8—The computer-implemented method of any one of clauses 1-7, wherein the object comprises an object that diffusely reflects light.

(166) Clause 9—The computer-implemented method of any one of clauses 1-8, wherein the computing device comprises a camera, and the method further including: generating the particular input image using the camera; and receiving, at the computing device, the generated particular input image from the camera.

(167) Clause 10—The computer-implemented method of any one of clauses 1-9, further including: providing the optimized direction of synthetic lighting using the computing device.

(168) Clause 11—The computer-implemented method of any one of clauses 1-10, wherein an illumination profile of the input image is estimated using a light estimation model, and wherein the method further including: providing a prediction of the original illumination profile using the

computing device.
(169) Clause 12—The computer-implemented method of clause 11, wherein the prediction of the illumination profile includes: generating a high dynamic range (HDR) lighting environment based on low dynamic range (LDR) images of a set of reference objects, each of the set of reference objects having a respective bidirectional reflectance distribution function (BRDF).
(170) Clause 13—The computer-implemented method of clause 12, wherein the set of reference objects includes one or more of a mirror ball, a matte silver ball, or a gray diffuse ball.
(171) Clause 14—The computer-implemented method of any one of clauses 1-13, wherein optimizing the direction of synthetic lighting by using the trained neural network comprises: obtaining the trained neural network at the computing device; and determining the direction of synthetic lighting by the computing device using the obtained neural network.
(172) Clause 15—The computer-implemented method of clause 14, wherein training the neural network comprises training the neural network at the computing device.
(173) Clause 16—The computer-implemented method of any one of clauses 1-15, further including: applying, by the computing device, a geometry model to the particular input image to determine a surface orientation map indicative of a distribution of lighting on the object in the particular input image based on a surface geometry of the object.
(174) Clause 17—The computer-implemented method of any one of clauses 1-16, further including: applying, by the computing device and based on a surface geometry of the object in the particular image and the optimized direction of synthetic lighting, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the particular input image.
(175) Clause 18—The computer-implemented method of clause 17, further including: enhancing, based on the quotient image, the portion of the particular input image.
(176) Clause 19—A computing device, including: one or more processors; and data storage, wherein the data storage has stored thereon computer-executable instructions that, when executed by the one or more processors, cause the computing device to carry out functions including the computer-implemented method of any one of clauses 1-18.
(177) Clause 20—An article of manufacture including one or more computer readable media having computer-readable instructions stored thereon that, when executed by one or more processors of a computing device, cause the computing device to carry out functions that comprise the computer-implemented method of any one of clauses 1-18.
(178) Clause 21—The article of manufacture of clause 20, wherein the one or more computer readable media comprise one or more non-transitory computer readable media.
(179) Clause 22—A computing device, including: means for carrying out the computer-implemented method of any one of clauses 1-18.
(180) Clause 23—A computer-implemented method, including: training a neural network to apply a geometry model to an input image to predict a surface orientation map indicative of a distribution of lighting on an object in the input image based on a surface geometry of the object; predicting, by a computing device, a surface orientation map by using the trained neural network to apply the geometry model to a particular object in a particular input image based on a particular surface geometry of the particular object; receiving, by the computing device, an indication of a direction of synthetic lighting to be applied to the particular input image to enhance at least a portion of the particular input image; and generating, by the computing device and based on the predicted surface orientation map and the indicated direction of synthetic lighting, a light visibility map for the particular input image, wherein the light visibility map is indicative of synthetic lighting to be applied to the particular input image based on the particular surface geometry of the particular object.
(181) Clause 24—The computer-implemented method of clause 23, further including: applying the generated light visibility map to the particular input image to enhance at least a portion of the

particular input image.

(182) Clause 25—The computer-implemented method of clause 24, wherein applying the generated light visibility map to the particular input image comprises: determining, by the computing device, a request to enhance the particular input image; sending the request to enhance the particular input image from the computing device to a second computing device, the second computing device including the trained neural network; and after sending the request, the computing device receiving, from the second computing device, an output image that applies the generated light visibility map to the particular input image.

(183) Clause 26—The computer-implemented method of any one of clauses 23-25, wherein training the neural network to apply the geometry model comprises training the neural network using a plurality of images of the object, where the plurality of images utilize a plurality of illumination profiles to light the object.

(184) Clause 27—The computer-implemented method of any one of clauses 23-26, wherein the neural network is a convolutional neural network.

(185) Clause 28—The computer-implemented method of any one of clauses 23-27, wherein the object comprises an object that diffusely reflects light.

(186) Clause 29—The computer-implemented method of any one of clauses 23-28, wherein the computing device comprises a camera, and the method further including: generating the particular input image of the object using the camera; and receiving, at the computing device, the generated particular input image from the camera.

(187) Clause 30—The computer-implemented method of any one of clauses 23-29, wherein receiving the indication of a direction of synthetic lighting includes: receiving, by the computing device, a user preference for the direction of synthetic lighting.

(188) Clause 31—The computer-implemented method of any one of clauses 23-29, wherein receiving the indication of a direction of synthetic lighting comprises: training a second neural network to apply an environmental light estimation model to the input image to optimize a direction of synthetic lighting; detecting, by the computing device, a pose of the particular object in the particular input image; and optimizing, by the computing device and based on the pose, the direction of synthetic lighting by using the trained second neural network to apply the environmental light estimation model to the particular input image.

(189) Clause 32—The computer-implemented method of any one of clauses 23-29, further including: providing the predicted surface orientation map using the computing device.

(190) Clause 33—The computer-implemented method of any one of clauses 23-32, wherein an illumination profile of the input image is modeled using an original light model, and wherein the method further including: providing a prediction of the original light model using the computing device.

(191) Clause 34—The computer-implemented method of clause 33, wherein the prediction of the illumination profile comprises: generating a high dynamic range (HDR) lighting environment based on low dynamic range (LDR) images of a set of reference objects, each of the set of reference objects having a respective bidirectional reflectance distribution function (BRDF).

(192) Clause 35—The computer-implemented method of clause 34, wherein the set of reference objects includes one or more of a mirror ball, a matte silver ball, or a gray diffuse ball.

(193) Clause 36—The computer-implemented method of any one of clauses 23-35, wherein predicting the surface orientation map by using the trained neural network comprises: obtaining the trained neural network at the computing device; and determining the surface orientation map by the computing device using the obtained neural network.

(194) Clause 37—The computer-implemented method of clause 36, wherein training the neural network comprises training the neural network at the computing device.

(195) Clause 38—The computer-implemented method of any one of clauses 23-37, further including: applying, by the computing device and based on the surface orientation map and the

direction of synthetic lighting, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the particular input image.
(196) Clause 39—The computer-implemented method of clause 38, further including: enhancing, based on the quotient image, the portion of the input image.
(197) Clause 40—A computing device, including: one or more processors; and data storage, wherein the data storage has stored thereon computer-executable instructions that, when executed by the one or more processors, cause the computing device to carry out functions including the computer-implemented method of any one of clauses 23-39.
(198) Clause 41—An article of manufacture including one or more computer readable media having computer-readable instructions stored thereon that, when executed by one or more processors of a computing device, cause the computing device to carry out functions that comprise the computer-implemented method of any one of clauses 23-39.
(199) Clause 42—The article of manufacture of clause 41, wherein the one or more computer readable media comprise one or more non-transitory computer readable media.
(200) Clause 43—A computing device, including: means for carrying out the computer-implemented method of any one of clauses 23-39.
(201) Clause 44—A computer-implemented method, including: training a neural network to apply a light energy model to an input image to predict a quotient image indicative of an amount of light energy to be applied to each pixel of the input image; receiving, by the computing device, a light visibility map for a particular input image, wherein the light visibility map is indicative of synthetic lighting to be applied to the particular input image based on a surface geometry of an object in the particular input image; and predicting, by a computing device, a quotient image by using the trained neural network to apply the light energy model to the particular input image.
(202) Clause 45—The computer-implemented method of clause 44, further including: applying the predicted quotient image to the particular input image to enhance at least a portion of the particular input image.
(203) Clause 46—The computer-implemented method of any one of clauses 44 or 45, further including: adjusting the quotient image to compensate for an exposure level in the particular input image.
(204) Clause 47—The computer-implemented method of any one of clauses 44-46, further including: adjusting the quotient image to compensate for a brightness level in the particular input image.
(205) Clause 48—The computer-implemented method of any one of clauses 44-47, further including: adjusting the quotient image to apply a matting refinement to the particular input image.
(206) Clause 49—The computer-implemented method of any one of clauses 45-49, wherein predicting quotient image comprises: determining, by the computing device, a request to enhance the particular input image; sending the request to enhance the particular input image from the computing device to a second computing device, the second computing device including the trained neural network; and after sending the request, the computing device receiving, from the second computing device, an output image that applies the predicted quotient image to the particular input image.
(207) Clause 50—The computer-implemented method of any one of clauses 44-49, wherein training the neural network to apply the light energy model comprises training the neural network using a plurality of images of the object, where the plurality of images utilize a plurality of illumination profiles to light the object.
(208) Clause 51—The computer-implemented method of any one of clauses 44-50, wherein the neural network is a convolutional neural network.
(209) Clause 52—The computer-implemented method of any one of clauses 44-51, wherein the object comprises an object that diffusely reflects light.
(210) Clause 53—The computer-implemented method of any one of clauses 44-52, wherein the

computing device comprises a camera, and the method further including: generating the particular input image of the object using the camera; and receiving, at the computing device, the generated particular input image from the camera.
(211) Clause 54—The computer-implemented method of any one of clauses 44-53, further including: receiving, by the computing device, a user preference for a direction of the synthetic lighting to be applied to the particular input image.
(212) Clause 55—The computer-implemented method of any one of clauses 44-54 further including: training a second neural network to apply an environmental light estimation model to the input image to optimize a direction of the synthetic lighting; detecting, by the computing device, a pose of the object in the particular input image; and optimizing, by the computing device and based on the pose, the direction of the synthetic lighting by using the trained second neural network to apply the environmental light estimation model to the particular input image.
(213) Clause 56—The computer-implemented method of any one of clauses 44-55, further including: applying, by the computing device, a geometry model to the particular input image to determine a surface orientation map indicative of a distribution of lighting on the object in the particular input image based on a surface geometry of the object, and wherein the light visibility map is based on the surface orientation map.
(214) Clause 57—The computer-implemented method of any one of clauses 44-56 wherein an illumination profile of the input image is modeled using an environmental light estimation model, and wherein the method further including: providing a prediction of the illumination profile using the computing device.
(215) Clause 58—The computer-implemented method of any one of clauses 44-57, wherein predicting the quotient image by using the trained neural network comprises: obtaining the trained neural network at the computing device; and determining the quotient image by the computing device using the obtained neural network.
(216) Clause 59—The computer-implemented method of clause 44-58, wherein training the neural network comprises training the neural network at the computing device.
(217) Clause 60—A computing device, including: one or more processors; and data storage, wherein the data storage has stored thereon computer-executable instructions that, when executed by the one or more processors, cause the computing device to carry out functions including the computer-implemented method of any one of clauses 44-59.
(218) Clause 61—An article of manufacture including one or more computer readable media having computer-readable instructions stored thereon that, when executed by one or more processors of a computing device, cause the computing device to carry out functions that comprise the computer-implemented method of any one of clauses 44-59.
(219) Clause 62—The article of manufacture of clause 61, wherein the one or more computer readable media comprise one or more non-transitory computer readable media.
(220) Clause 63—A computing device, including: means for carrying out the computer-implemented method of any one of clauses 44-59.
(221) Clause 64—A computer-implemented method, including: applying, by a computing device, a geometry model to an input image to determine a surface orientation map indicative of a distribution of lighting on an object in the input image based on a surface geometry of the object; applying, by the computing device, an environmental light estimation model to the input image to determine an optimal direction of synthetic lighting to be applied to the input image to enhance at least a portion of the input image; applying, by the computing device and based on the surface orientation map and the optimal direction of synthetic lighting, a light energy model to determine a quotient image indicative of an amount of light energy to be applied to each pixel of the input image; and enhancing, based on the quotient image, the portion of the input image.
(222) Clause 65—The computer-implemented method of clause 64, wherein applying the geometry model comprises: training a neural network to apply the geometry model to predict the surface

orientation map; and predicting the surface orientation map by using the trained neural network.
(223) Clause 66—The computer-implemented method of any one of clauses 64 or 65, wherein applying the environmental light estimation model comprises: training a neural network to apply the environmental light estimation model to optimize the direction of the synthetic lighting; and optimizing, by the computing device, the direction of the synthetic lighting by using the trained neural network.
(224) Clause 67—The computer-implemented method of any one of clauses 64-66, wherein applying the light energy model comprises: training a neural network to apply the light energy model to predict the quotient image; and predicting the quotient by using the trained neural network.
(225) Clause 68—The computer-implemented method of any one of clauses 64-67, further including: generating, by the computing device and based on the surface orientation map and the direction of synthetic lighting, a light visibility map, and wherein the quotient image is based on the light visibility map.
(226) Clause 69—The computer-implemented method of any one of clauses 64-68, wherein enhancing the portion of the input image comprises: determining, by the computing device, a request to enhance the input image; sending the request to enhance the input image from the computing device to a second computing device, the second computing device including the trained neural network; and after sending the request, the computing device receiving, from the second computing device, an output image that applies the quotient image to enhance the portion of the input image.
(227) Clause 70—The computer-implemented method of any one of clauses 65-68, further including: training the neural network using a plurality of images of the object, where the plurality of images utilize a plurality of illumination profiles to light the object.
(228) Clause 71—The computer-implemented method of any one of clauses 65-68, or 70, wherein the neural network is a convolutional neural network.
(229) Clause 72—The computer-implemented method of any one of clauses 64-71, wherein the object comprises an object that diffusely reflects light.
(230) Clause 73—The computer-implemented method of any one of clauses 64-72, wherein the computing device comprises a camera, and the method further including: generating the input image of the object using the camera; and receiving, at the computing device, the generated input image from the camera.
(231) Clause 74—The computer-implemented method of any one of clauses 64-73, further including: providing the enhanced input image using the computing device.
(232) Clause 75—The computer-implemented method of any one of clauses 64-74, wherein an illumination profile of the input image is modeled using an environmental light estimation model, and wherein the method further including: providing a prediction of the illumination profile using the computing device.
(233) Clause 76—The computer-implemented method of clause 75, wherein the prediction of the illumination profile includes: generating a high dynamic range (HDR) lighting environment based on low dynamic range (LDR) images of a set of reference objects, each of the set of reference objects having a respective bidirectional reflectance distribution function (BRDF).
(234) Clause 77—The computer-implemented method of clause 76, wherein the set of reference objects includes one or more of a mirror ball, a matte silver ball, or a gray diffuse ball.
(235) Clause 78—The computer-implemented method of any one of clauses 65-67 or 70-71, wherein the predicting using the trained neural network comprises: obtaining the trained neural network at the computing device; and applying the obtained neural network at the computing device.
(236) Clause 79—The computer-implemented method of clause 65-66 or 70-71, wherein training the neural network comprises training the neural network at the computing device.

(237) Clause 80—A computing device, including: one or more processors; and data storage, wherein the data storage has stored thereon computer-executable instructions that, when executed by the one or more processors, cause the computing device to carry out functions including the computer-implemented method of any one of clauses 64-79.
(238) Clause 81—An article of manufacture including one or more computer readable media having computer-readable instructions stored thereon that, when executed by one or more processors of a computing device, cause the computing device to carry out functions that comprise the computer-implemented method of any one of clauses 64-79.
(239) Clause 82—The article of manufacture of clause 81, wherein the one or more computer readable media comprise one or more non-transitory computer readable media.
(240) Clause 83—A computing device, including: means for carrying out the computer-implemented method of any one of clauses 64-79.
(241) The present disclosure is not to be limited in terms of the particular embodiments described in this application, which are intended as illustrations of various aspects. Many modifications and variations can be made without departing from its spirit and scope, as will be apparent to those skilled in the art. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those enumerated herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims.
(242) The above detailed description describes various features and functions of the disclosed systems, devices, and methods with reference to the accompanying figures. In the figures, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, figures, and claims are not meant to be limiting. Other embodiments can be utilized, and other changes can be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.
(243) With respect to any or all of the ladder diagrams, scenarios, and flow charts in the figures and as discussed herein, each block and/or communication may represent a processing of information and/or a transmission of information in accordance with example embodiments. Alternative embodiments are included within the scope of these example embodiments. In these alternative embodiments, for example, functions described as blocks, transmissions, communications, requests, responses, and/or messages may be executed out of order from that shown or discussed, including substantially concurrent or in reverse order, depending on the functionality involved. Further, more or fewer blocks and/or functions may be used with any of the ladder diagrams, scenarios, and flow charts discussed herein, and these ladder diagrams, scenarios, and flow charts may be combined with one another, in part or in whole.
(244) A block that represents a processing of information may correspond to circuitry that can be configured to perform the specific logical functions of a herein-described method or technique. Alternatively or additionally, a block that represents a processing of information may correspond to a module, a segment, or a portion of program code (including related data). The program code may include one or more instructions executable by a processor for implementing specific logical functions or actions in the method or technique. The program code and/or related data may be stored on any type of computer readable medium such as a storage device including a disk or hard drive or other storage medium.
(245) The computer readable medium may also include non-transitory computer readable media such as non-transitory computer-readable media that stores data for short periods of time like register memory, processor cache, and random access memory (RAM). The computer readable media may also include non-transitory computer readable media that stores program code and/or

data for longer periods of time, such as secondary or persistent long term storage, like read only memory (ROM), optical or magnetic disks, compact-disc read only memory (CD-ROM), for example. The computer readable media may also be any other volatile or non-volatile storage systems. A computer readable medium may be considered a computer readable storage medium, for example, or a tangible storage device.

(246) Moreover, a block that represents one or more information transmissions may correspond to information transmissions between software and/or hardware modules in the same physical device. However, other information transmissions may be between software modules and/or hardware modules in different physical devices.

(247) While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are provided for explanatory purposes and are not intended to be limiting, with the true scope being indicated by the following claims.

Claims

1. A computer-implemented method, comprising: determining, by a geometry model of a computing device, a surface orientation map based on a prediction of lighting on an object in an input image based on a surface geometry of the object; determining, by an environmental light estimation model of the computing device, a direction of synthetic lighting to be applied to the input image to modify at least a portion of the input image; determining, by a light energy model of the computing device and based on the surface orientation map and the direction of synthetic lighting, a quotient image indicative of an amount of light energy to be applied to each pixel of the input image; and modifying, based on a product of the quotient image and the input image, the portion of the input image.
2. The computer-implemented method of claim 1, wherein the determining of the direction of synthetic lighting comprises: detecting, by the computing device, a pose of the object in the input image, and wherein the determining of the direction of the synthetic lighting is based on the pose.
3. The computer-implemented method of claim 1, further comprising: generating, by the computing device and based on the surface orientation map and the direction of synthetic lighting, a light visibility map, and wherein the determining of the quotient image is based on the light visibility map.
4. The computer-implemented method of claim 1, wherein the modifying of the portion of the input image comprises: determining, by the computing device, a request to modify the portion of the input image; sending the request to request to modify the portion of the input image from the computing device to a second computing device, the second computing device comprising a trained neural network; and after sending the request, the computing device receiving, from the second computing device, an output image that applies the quotient image to modify the portion of the input image.
5. The computer-implemented method of claim 1, wherein the determining of the direction of synthetic lighting comprises: receiving, by the computing device, a user preference for the direction of the synthetic lighting to be applied to the input image.
6. The computer-implemented method of claim 1, wherein the object has a characteristic of diffusely reflecting light.
7. The computer-implemented method of claim 1, wherein the computing device comprises a camera, and the method further comprising: generating the input image of the object using the camera; and receiving, at the computing device, the generated input image from the camera.
8. The computer-implemented method of claim 1, further comprising: providing the portion of the modified input image using the computing device.
9. The computer-implemented method of claim 1, further comprising: predicting an illumination

profile of the input image using the environmental light estimation model; and providing the predicted illumination profile using the computing device.

10. The computer-implemented method of claim 9, wherein the prediction of the illumination profile comprises: generating a high dynamic range (HDR) lighting environment based on low dynamic range (LDR) images of a plurality of reference objects, each of the reference objects having a respective bidirectional reflectance distribution function (BRDF).

11. The computer-implemented method of claim 10, wherein the set of reference objects includes one or more of a mirror ball, a matte silver ball, or a gray diffuse ball.

12. The computer-implemented method of claim 1, further comprising: adjusting the quotient image to apply one or more of: (i) a first compensation for an exposure level in the input image, (ii) a second compensation for a brightness level in the input image, or (iii) a matting refinement to the input image.

13. A computing device, comprising: one or more processors; and data storage, wherein the data storage has stored thereon computer-executable instructions that, when executed by the one or more processors, cause the computing device to carry out functions comprising: determining, by a geometry model of the computing device, a surface orientation map based on a prediction of lighting on an object in an input image based on a surface geometry of the object; determining, by an environmental light estimation model of the computing device, a direction of synthetic lighting to be applied to the input image to modify at least a portion of the input image; determining, by a light energy model of the computing device and based on the surface orientation map and the direction of synthetic lighting, a quotient image indicative of an amount of light energy to be applied to each pixel of the input image; and modifying, based on a product of the quotient image and the input image, the portion of the input image.

14. An article of manufacture comprising one or more computer readable media having computer-readable instructions stored thereon that, when executed by one or more processors of a computing device, cause the computing device to carry out functions comprising: determining, by a geometry model of the computing device, a surface orientation map based on a prediction of lighting on an object in an input image based on a surface geometry of the object; determining, by an environmental light estimation model of the computing device, a direction of synthetic lighting to be applied to the input image to modify at least a portion of the input image; determining, by a light energy model of the computing device and based on the surface orientation map and the direction of synthetic lighting, a quotient image indicative of an amount of light energy to be applied to each pixel of the input image; and modifying, based on a product of the quotient image and the input image, the portion of the input image.

15. The computer-implemented method of claim 1, wherein at least a part of one or more of the geometry model, the environmental light estimation model, or the light energy model is part of a machine learning model.

16. The computer-implemented method of claim 15, further comprising: training the machine learning model to perform one or more of: (1) predict the surface orientation map, (2) predict the direction of the synthetic lighting, or (3) predict the quotient image.

17. The computer-implemented method of claim 15, wherein the machine learning model is trained to predict the surface orientation map, and the method further comprising: predicting the surface orientation map by using the trained machine learning model.

18. The computer-implemented method of claim 15, wherein the machine learning model is trained to predict the direction of the synthetic lighting, and the method further comprising: predicting the direction of the synthetic lighting by using the trained machine learning model.

19. The computer-implemented method of claim 15, wherein the machine learning model is trained to predict the quotient image, and the method further comprising: predicting the quotient image by using the trained machine learning model.

20. The computer-implemented method of claim 15, wherein the training of the machine learning

model is based on a training dataset comprising a plurality of images of the object, wherein the plurality of images utilize a plurality of illumination profiles to illuminate the object.

21. The computer-implemented method of claim 15, wherein the training of the machine learning model comprises training the machine learning model at the computing device.